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Numerical Modeling of Electromagnetic Scattering for Radar Cross Section Prediction: A Computational Comparison of Conventional and Stealth Aircraft Geometries

A Project Based Learning(PBL) Lab Report

Avionics

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Department of Aerospace Engineering



CERTIFICATE

Certified that the Project Based Learning (PBL) report titled ***Numerical Modeling of Electromagnetic Scattering for Radar Cross Section Prediction: A Computational Comparison of Conventional and Stealth Aircraft Geometries*** is carried out by **Akula Uday Kiran (1RV23AS004)**, **Shanthosh K V (1RV23AS053)**, and **Tejas L (1RV23AS061)**, who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfilment of the requirements for the degree of **Bachelor of Engineering in Aerospace Engineering** of the Visvesvaraya Technological University, Belagavi, during the year 2025–26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the project report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed by the institution for the said degree.

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Further, we declare that the content of this report has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru, and we will be one of the authors of the same.

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ABSTRACT

This report presents a first-principles computational framework for predicting the Radar Cross Section (RCS) of aerospace geometries and applies it to a comparative study of a conventional aircraft configuration against a shaped, low-observable “stealth” configuration. Radar Cross Section quantifies how effectively a target reflects electromagnetic energy back toward a radar receiver and, through the fourth-root dependence of detection range on RCS, is one of the most influential parameters in modern aircraft survivability design. The framework integrates Physical Optics (PO) for specular reflection, the Physical Theory of Diffraction (PTD) for edge diffraction, and second- and third-order ray tracing for double- and triple-bounce corner-reflector mechanisms, all summed coherently as complex amplitudes before the final magnitude-squared RCS conversion.

The Physical Optics solver is validated against the exact Mie series solution for a conducting sphere, matching to within 0.2 dBsm. Six triangulated-mesh geometries – a sphere, a flat plate, a conventional aircraft, a stealth aircraft, a flying wing, and a chined-nose fighter – are analysed across azimuth, elevation, and frequency, and the scattering mechanisms are decomposed feature-by-feature to identify which geometric elements drive which physical mechanism. The stealth aircraft is shown to achieve an 11.9 dB RCS reduction over the conventional aircraft at the nose-on aspect, equivalent to a radar detection-range reduction from 200 km to 100 km.

The framework is further extended with bistatic and polarimetric RCS formulations, a Surface Impedance Boundary Condition (SIBC) for finite-conductivity real aerospace materials (aluminium, titanium, stainless steel, and carbon-fibre-reinforced polymer) in place of the idealised Perfect Electric Conductor assumption, a gradient-free differential-evolution optimizer that searches the stealth geometry’s sweep, cant, and nose-length parameters for a minimum mean RCS, a wideband stepped-frequency impulse-response processor that resolves individual scattering centres along the airframe, and an Inverse Synthetic Aperture Radar (ISAR) imaging processor that forms a synthetic range/cross-range image of each aircraft from a coherent frequency-and-angle sweep. Every extension is accompanied by its own mesh-convergence study, and results that did not fully converge within practical computational limits (notably the $O(n^3)$ triple-bounce solver) are reported honestly rather than masked.

The report is organized into five chapters covering the introduction, theoretical background, methodology, results and discussion, and conclusions with future work, followed by an appendix of analytical calculation details. The expected outcome is a validated, physically transparent computational pipeline that quantitatively connects specific stealth-shaping design principles – leading-edge sweep, vertical-surface cant, and edge alignment – to their measured effect on radar cross section.

Keywords: Radar Cross Section, Physical Optics, Physical Theory of Diffraction, stealth aircraft, Mie series validation, double-bounce, triple-bounce, bistatic radar, polarimetric radar, Surface Impedance Boundary Condition, differential evolution, wideband impulse response, Inverse Synthetic Aperture Radar.



NOMENCLATURE

Roman Symbols

A_i	Area of the i -th triangular facet	[m ²]
B	Frequency bandwidth	[Hz]
c	Speed of light in free space	[m s ⁻¹]
$\mathbf{c}_i$	Centroid position vector of the i -th facet	[m]
$D(n, \beta)$	Ufimtsev fringe-wave diffraction coefficient	[-]
E_i, E_s	Incident and scattered electric field	[V m ⁻¹]
f	Radar frequency	[Hz]
f_{PO}, f_{PTD}, f_{DB}	Coherent complex scatter amplitude, PO/PTD/double-bounce	[m]
G_t, G_r	Transmit and receive antenna gain	[-]
$h(t)$	Time-domain range-profile impulse response	[-]
$H(f)$	Wideband coherent frequency response	[m]
k	Free-space wavenumber, $2\pi/\lambda$	[rad m ⁻¹]
L	Characteristic edge length of a corner reflector	[m]
n	PTD wedge interior-angle parameter	[-]
$\hat{n}_i$	Outward unit normal of the i -th facet	[-]
N	Number of facets / sample points	[-]
p, q	Polarisation indices (H or V)	[-]
P_t	Transmitted power	[W]
R	Range from radar to target	[m]
R_{max}	Maximum radar detection range	[m]
S	Polarimetric scattering matrix	[-]
S_{min}	Minimum detectable signal power	[W]
$\hat{s}$	Unit look/illumination direction vector	[-]
$W(f)$	Frequency-domain window function	[-]
Z_0	Free-space wave impedance (377Ω)	[Ω]
Z_s	Surface impedance (Leontovich SIBC)	[Ω]

Greek Symbols

β	Incidence angle from an edge (PTD)	[°]
γ	V-tail cant angle	[°]
Γ	Reflection coefficient	[-]
δ	Skin depth	[m]
ΔR	Range resolution	[m]
ΔCR	Cross-range resolution	[m]
$\Delta\theta$	ISAR angular aperture	[rad]
θ	Azimuth aspect angle	[°]
Λ	Wing leading-edge sweep angle	[°]
λ	Radar wavelength	[m]
μ	Magnetic permeability	[H m ⁻¹]
ρ_i	Reflectivity of the i -th facet (RAM/SIBC scalar)	[-]
σ	Radar cross section (RCS)	[m ²]
σ_{pq}	Polarimetric RCS, channel pq	[m ²]
$\sigma_{lin}, \sigma_{dB}$	Mean RCS, linear and dB domain	[m ²],[dBsm]

ABBREVIATIONS

DE	Differential Evolution
dBsm	Decibels relative to one square metre
FFT/IFFT	(Inverse) Fast Fourier Transform
GTD	Geometrical Theory of Diffraction
HH/VV/HV/VH	Polarimetric channel notation (transmit–receive)
ISAR	Inverse Synthetic Aperture Radar
MoM	Method of Moments
PEC	Perfect Electric Conductor
PO	Physical Optics
PTD	Physical Theory of Diffraction
RAM	Radar-Absorbent Material
RCS	Radar Cross Section
RVCE	RV College of Engineering
SAR	Synthetic Aperture Radar
SIBC	Surface Impedance Boundary Condition
TE/TM	Transverse Electric / Transverse Magnetic
VTU	Visvesvaraya Technological University

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Chapter 1

Introduction

CHAPTER 1

INTRODUCTION

Radar remains the primary means by which modern air-defence and surveillance systems detect, track, and identify aerospace vehicles. The Radar Cross Section (RCS) of an aircraft – the equivalent area of a hypothetical isotropic reflector that would return the same echo power as the real target – is the single electromagnetic parameter that governs how visible that aircraft is to a given radar, independent of its physical size. RCS is not a fixed number: it depends strongly on the aircraft’s external geometry, its orientation relative to the radar, the radar’s operating frequency, and, to a lesser extent, the electrical properties of the airframe skin material. Understanding and controlling this dependence is the foundation of low-observable, or “stealth”, aircraft design.

Aircraft external geometry is historically shaped first for aerodynamic performance, structural load paths, and internal volume, with radar visibility addressed afterward through applied coatings. Low-observable design instead treats radar return as a first-class shaping constraint: flat panels are angled away from likely threat directions, edges are swept to a small number of common angles so their diffraction lobes point away from the radar, and right-angle corner reflectors formed by wing-fuselage or fin-fuselage junctions are eliminated or canted. Reproducing and quantifying this shaping effect computationally, from first principles, is the central task undertaken in this report.

1.1 Background and Motivation

Electromagnetic scattering from an aircraft-scale target is, in general, only solvable exactly using full-wave numerical methods such as the Method of Moments (MoM), which require of the order of ten unknowns per wavelength of surface area and become computationally intractable for an electrically large target such as a full-size aircraft at microwave frequencies. High-frequency asymptotic methods – Physical Optics (PO) for specular reflection and the Physical Theory of Diffraction (PTD) for edge diffraction – instead approximate each small facet of a triangulated surface mesh as behaving locally like an infinite flat plate or straight edge, respectively. These methods trade exactness for tractability and are the same methods used, in various forms, by production RCS-prediction codes such as XPATCH.

Real aircraft return energy through several simultaneous, physically distinct mechanisms: specular glint off flat or gently curved panels, diffraction that “leaks” around sharp edges, and second- or third-order re-reflection inside corner-reflector-like junctions between panels. These mechanisms must be summed as complex, phase-carrying amplitudes – not as independent powers – because they can constructively or destructively interfere depending on the relative path length of each contribution. This coherent-summation requirement, together with the

strongly non-linear dependence of RCS on aspect angle, edge wedge angle, and dihedral/trihedral corner geometry, motivates the layered computational framework developed in this report.

1.2 Problem Statement

The problem addressed in this study is to develop, validate, and apply a first-principles computational framework that predicts the monostatic and bistatic Radar Cross Section of faceted aerospace geometries, to use that framework to quantify the RCS reduction achieved by low-observable shaping relative to a conventional aircraft configuration, and to attribute the total RCS at any given aspect angle to specific, named physical scattering mechanisms and specific geometric features.

Three concrete challenges are addressed. First, three physically distinct scattering mechanisms – specular reflection, edge diffraction, and multi-bounce re-reflection – must be modelled coherently, with phase, rather than combined as independent, incoherent contributions. Second, RCS depends non-linearly on surface orientation, edge wedge angle, and dihedral/trihedral corner geometry, so a coarse angular or geometric sampling can hide unfavourable behaviour. Third, which geometric feature drives which mechanism, and by how much, must be extracted quantitatively from the computed results rather than assumed from visual inspection of the geometry.

1.3 Research Gap

Physical Optics and the Physical Theory of Diffraction are individually well established for canonical scattering problems – flat plates, cylinders, wedges, and spheres – and are documented extensively in the classical RCS literature. Corner-reflector (dihedral and trihedral) retroreflection is likewise well understood as an isolated canonical problem, being the basis of deliberately-designed radar calibration targets. However, a single, internally consistent computational framework that combines all three mechanisms coherently over a complete faceted aircraft geometry, extends them to bistatic and polarimetric operation, incorporates a realistic finite-conductivity surface-impedance boundary condition in place of the idealised Perfect Electric Conductor assumption, and couples the result to an automated shape optimizer and to wideband/ISAR imaging post-processing, is not commonly presented as a single teaching-scale, first-principles Python implementation with an accompanying mesh-convergence discipline at every stage. The present report addresses this gap.

1.4 Objectives

The primary objective of this project is to design, implement, and validate a first-principles Radar Cross Section computational framework and to use it to quantify how geometric shaping reduces

the radar visibility of a stealth aircraft configuration relative to a conventional configuration.

The secondary objectives are as follows:

1. To review the theoretical basis of Physical Optics, the Physical Theory of Diffraction, multi-bounce ray tracing, and Radar-Absorbent Material (RAM) modelling.
2. To validate the Physical Optics solver against the exact analytical Mie series solution for a conducting sphere.
3. To analyse the RCS of six triangulated-mesh geometries across azimuth, elevation, and frequency, and to decompose the total RCS into its constituent physical mechanisms.
4. To extend the framework to bistatic and polarimetric operation, and to a Surface Impedance Boundary Condition representing real, finite-conductivity aerospace materials.
5. To formulate and solve a gradient-free geometry-optimization problem that searches for the stealth-shaping parameter set that minimises mean RCS.
6. To implement wideband stepped-frequency impulse-response and Inverse Synthetic Aperture Radar (ISAR) imaging post-processing on top of the validated coherent scattering framework.
7. To perform a mesh-convergence study for every result presented in this report, and to report honestly any result that does not fully converge within practical computational limits.

1.5 Scope of the Study

The study is limited to faceted, triangulated-mesh representations of six aerospace geometries – a sphere, a flat plate, a conventional aircraft, a stealth aircraft, a flying wing, and a chined-nose fighter – analysed using high-frequency asymptotic electromagnetic methods (Physical Optics, the Physical Theory of Diffraction, and geometric-optics ray tracing for multi-bounce mechanisms) rather than full-wave numerical solution of Maxwell's equations. All geometries are treated as Perfect Electric Conductors except where the Surface Impedance Boundary Condition extension is explicitly applied. The computational framework and every result in this report were implemented and generated in Python, using NumPy, SciPy, and Matplotlib, without recourse to a commercial electromagnetic solver.



Chapter 2

Theoretical Background

CHAPTER 2

THEORETICAL BACKGROUND

This chapter presents the theoretical background required for the computational Radar Cross Section (RCS) framework developed in this report. The equations are arranged in the same sequence followed by the computational pipeline: the RCS definition and radar range equation motivate why RCS is studied at all, the facet representation and Physical Optics (PO) formulation provide the dominant specular-reflection mechanism, the Physical Theory of Diffraction (PTD) adds edge diffraction, geometric-optics ray tracing adds double- and triple-bounce corner-reflector mechanisms, the bistatic and polarimetric generalisations extend every mechanism to arbitrary transmit/receive geometry and polarisation, the Radar-Absorbent Material (RAM) and Surface Impedance Boundary Condition (SIBC) formulations replace the idealised Perfect Electric Conductor (PEC) assumption with realistic lossy surfaces, and the geometry-optimization, wideband impulse-response, and ISAR-imaging formulations define the post-processing built on top of the validated coherent scattering framework. A bracketed source note is provided immediately after each equation to link the equation to its textbook or journal reference.

2.1 Overall Theoretical Flow

The analysis begins with the RCS definition, proceeds through the coherent-summation electromagnetic scattering mechanisms (PO, PTD, double-bounce, triple-bounce), and then branches into the extension formulations built on top of that coherent complex amplitude. This order is used so that each stage supplies the input required by the next.

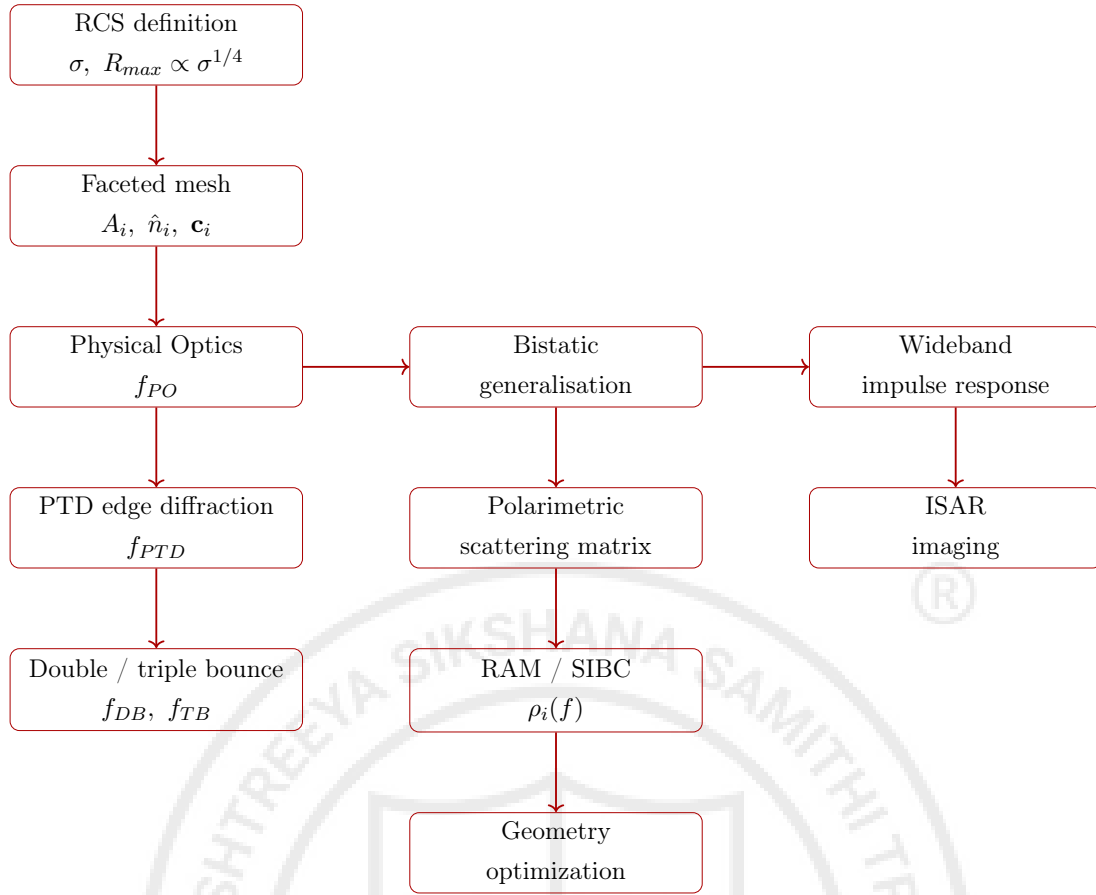


Figure 2.1: Logical connection between the theoretical formulations used in the RCS computational framework.

2.2 Radar Cross Section Definition and Radar Range Equation

Radar Cross Section is defined as the limiting value, as observation range tends to infinity, of the scaled ratio of scattered to incident power density:

$$\sigma = \lim_{R \rightarrow \infty} 4\pi R^2 \frac{|E_s|^2}{|E_i|^2}. \quad (1)$$

(Source: RCS definition from Knott, Shaeffer and Tuley [3].)

Here E_i and E_s are the incident and scattered electric fields and R is the range to the observation point. Substituting σ into the monostatic radar range equation and solving for the maximum detection range gives

$$R_{max} = \left(\frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 S_{min}} \right)^{1/4} \implies R_{max} \propto \sigma^{1/4}, \quad (2)$$

(Source: Radar range equation from Skolnik [4].)

where P_t is transmitted power, G_t and G_r are the transmit and receive antenna gains, λ is

wavelength, and S_{min} is the minimum detectable signal power. The fourth-root dependence in Eq. (2) is the central practical motivation for this report: an order-of-magnitude reduction in RCS reduces detection range only by a factor of $10^{1/4} \approx 1.78$, so RCS reduction through shaping is a far more efficient lever on detectability than increasing radar transmit power is on detection range.

2.3 Facet Representation and Physical Optics

The aircraft surface is discretised into flat triangular facets, each carrying a centroid position $\mathbf{c}_i$, an outward unit normal $\hat{\mathbf{n}}_i$, and an area A_i . A facet is illuminated when $\hat{\mathbf{n}}_i \cdot \hat{\mathbf{s}} > 0$, where $\hat{\mathbf{s}}$ is the unit look direction from the target to the radar. The coherent Physical Optics scattered-field amplitude is the phased sum over all illuminated facets:

$$f_{PO} = \sum_i \rho_i A_i (\hat{\mathbf{n}}_i \cdot \hat{\mathbf{s}}) \exp(j2k \mathbf{c}_i \cdot \hat{\mathbf{s}}), \quad \sigma = \frac{4\pi}{\lambda^2} |f|^2, \quad (3)$$

(Source: Physical Optics facet-sum formulation from Knott, Shaeffer and Tuley [3].)

where $k = 2\pi/\lambda$ is the free-space wavenumber and ρ_i is the per-facet reflectivity ($\rho_i = 1$ for bare PEC, $\rho_i < 1$ under a RAM/SIBC model as introduced in Section 2.10). The PO return scales with facet area and incidence angle as

$$\sigma \propto \frac{A^2 (\hat{\mathbf{n}} \cdot \hat{\mathbf{s}})^2}{\lambda^2}, \quad (4)$$

(Source: Physical Optics area-scaling relation from Ruck *et al.* [9].)

so that large flat panels facing the radar dominate the total return; this quadratic-in-area law is the physical basis for splitting large flat panels into several smaller, differently-angled facets during stealth shaping.

2.4 Physical Theory of Diffraction

Physical Optics is only valid in the interior of a flat facet; at an edge, the surface is discontinuous and the PO prediction is not merely imprecise but formally discontinuous across the shadow boundary. The Physical Theory of Diffraction adds a correction “fringe wave” localised to the edge so that the combined PO+PTD field is continuous everywhere. The Ufimtsev fringe-wave diffraction coefficient for a wedge of interior-angle parameter n and incidence angle β measured from the edge is

$$D(n, \beta) = -\frac{e^{-j\pi/4}}{n\sqrt{2\pi k}} \cdot \frac{\cot(\pi/2n)}{\sin \beta}, \quad (5)$$

(Source: Ufimtsev fringe-wave diffraction coefficient from Ufimtsev [2], following the Geometrical Theory

of Diffraction of Keller [7] and its uniform reformulation by Kouyoumjian and Pathak [8].)

where $n = 2$ for a free (boundary) edge and $n = 1.5$ for a right-angle convex internal ridge. Diffraction lobes from a finite edge of length L follow a sinc envelope in aspect angle:

$$f_{PTD} \propto \text{sinc}(kL \cos \beta). \quad (6)$$

(Source: Finite-edge diffraction-lobe relation from Ufimtsev [2].)

2.5 Double-Bounce Corner-Reflector Condition

Two facets meeting at close to a right angle form a dihedral corner reflector that retroreflects incident energy back along its incoming path over a wide range of incidence angles. A facet pair (i, j) is treated as an active double-bounce contributor when the retroreflection condition

$$\mathbf{d}_{ref,j} \cdot \hat{s} > 0.2 \quad (7)$$

(Source: Retroreflection activation condition for faceted double-bounce ray tracing, following the corner-reflector formulation of Jenn [10].)

is satisfied, where $\mathbf{d}_{ref,j}$ is the ray reflected first off facet i and then off facet j . Because a dihedral corner stays retroreflective over a much wider range of incidence angle than a single flat plate does, double-bounce contributions appear as sharp but comparatively broad RCS spikes at quarter-on aspect angles.

2.6 Triple-Bounce Trihedral Corner Reflector

Extending the ray trace one further reflection captures trihedral corners – three mutually near-perpendicular facets meeting at a point, the same geometry deliberately used in trihedral radar calibration reflectors. The classical peak RCS of a trihedral corner reflector with characteristic edge length L is

$$\sigma_{trihedral, peak} \sim \frac{4\pi L^4}{3\lambda^2}, \quad (8)$$

(Source: Trihedral corner-reflector peak RCS relation from the Radar Cross Section Handbook of Ruck *et al.* [9].)

which scales with the fourth power of edge length, a stronger scaling than the simple dihedral of Section 2.5, and explains why triple-bounce peak returns are found numerically to exceed double-bounce peak returns at their respective optimum aspect angles.

2.7 Bistatic Generalisation

Every mechanism above generalises to independent transmitter and receiver directions, $\hat{s}_{tx}$ and $\hat{s}_{rx}$, via the incidence–scatter bisector:

$$\text{bisector} = \frac{1}{2}(\hat{s}_{tx} + \hat{s}_{rx}), \quad \text{obliquity} = \hat{n} \cdot \text{bisector}, \quad \text{phase} = \exp(jk \mathbf{c} \cdot (\hat{s}_{tx} + \hat{s}_{rx})). \quad (9)$$

(Source: Bistatic tangent-plane Physical Optics generalisation, following the bistatic scattering formulation of Balanis [5].)

At $\hat{s}_{rx} = \hat{s}_{tx}$, Eq. (9) reduces exactly to the monostatic formulation of Sections 2.3–2.6, a property verified numerically to floating-point precision in the implemented framework.

2.8 Polarimetric Scattering Matrix

A full 2×2 complex scattering matrix relates the incident and scattered fields in a fixed global horizontal/vertical (H,V) polarisation basis:

$$\begin{bmatrix} E_H^s \\ E_V^s \end{bmatrix} = S \begin{bmatrix} E_H^i \\ E_V^i \end{bmatrix}, \quad \sigma_{pq} = \frac{4\pi}{\lambda^2} |S_{pq}|^2, \quad p, q \in \{H, V\}. \quad (10)$$

(Source: Polarimetric scattering-matrix formulation from Balanis [5].)

Each PEC facet reflects with the transverse-electric/transverse-magnetic (TE/TM) reflection dichotomy ($r_{\perp} = -1$, $r_{\parallel} = +1$) in its own local plane-of-incidence basis; projecting that local reflection into the fixed global frame is what produces cross-polarised return (σ_{HV}, σ_{VH}). Multi-bounce mechanisms use cascaded Jones matrices, for example double-bounce $S_{DB} \propto J_j J_i$ and triple-bounce $S_{TB} \propto J_k J_j J_i$.

2.9 Radar-Absorbent Material: Salisbury Screen and Dallenbach Layer

Radar-Absorbent Material coatings convert incident energy into heat instead of reflecting it, and are used where geometric shaping cannot redirect a surface away from every threat direction. Both classical designs are transmission-line reflection problems characterised by an input impedance Z_{in} and reflection coefficient

$$\Gamma = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}, \quad Z_0 = 377 \Omega \text{ (free-space impedance)}. \quad (11)$$

(Source: Transmission-line reflection-coefficient formulation from Balanis [5].)

A **Salisbury screen** places a single resistive sheet ($R_s \approx Z_0$) a quarter-wavelength in front of a PEC backplane, giving deep but narrowband absorption at the design frequency. A **Dallenbach layer** instead uses a single lossy dielectric slab of complex permittivity ϵ_r and thickness d directly on the PEC backplane:

$$Z_{in} = Z_1 \tanh(jk_1 d), \quad Z_1 = \frac{Z_0}{\sqrt{\epsilon_r \mu_r}}, \quad k_1 = k_0 \sqrt{\epsilon_r \mu_r}, \quad (12)$$

(Source: Dallenbach absorbing-layer impedance formulation from Balanis [5].)

which trades peak absorption depth for a broader useful bandwidth than the Salisbury design.

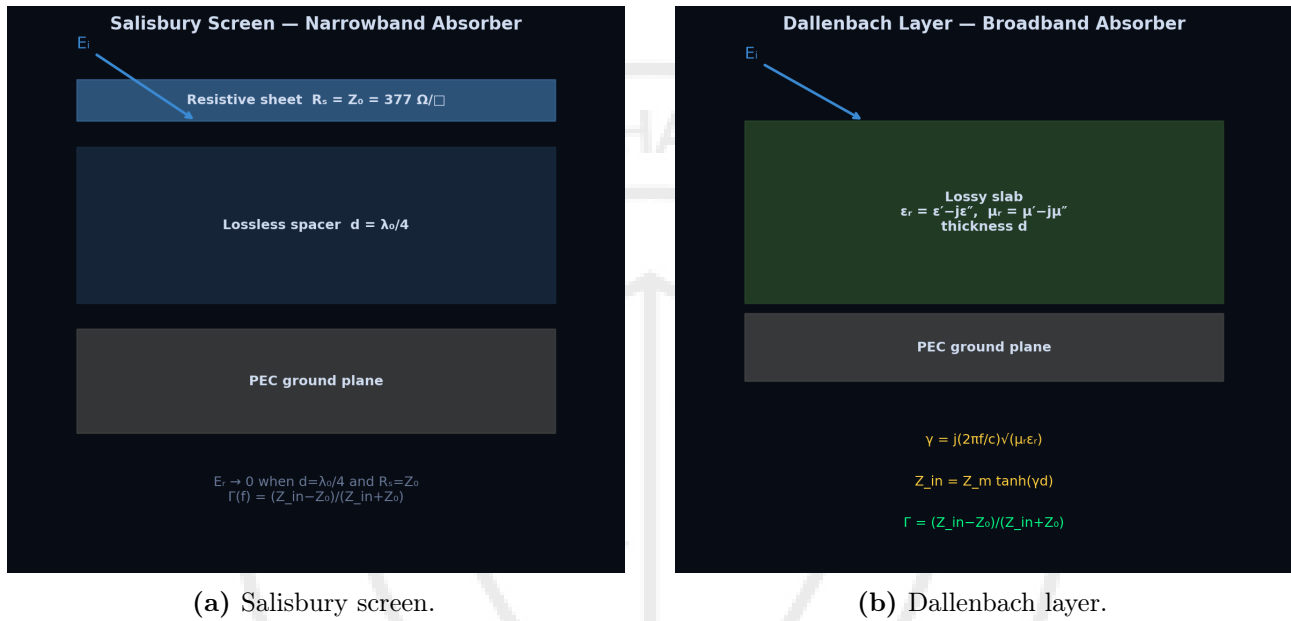


Figure 2.2: Radar-Absorbent Material configurations modelled in this report: the Salisbury screen gives deep narrowband absorption, the Dallenbach layer gives broader shallower absorption.

2.10 Surface Impedance Boundary Condition

Every mechanism above assumes a Perfect Electric Conductor. Real airframe skins are finite-conductivity materials, and the Leontovich Surface Impedance Boundary Condition (SIBC) replaces the PEC assumption with a frequency-dependent surface impedance:

$$Z_s = (1 + j) \sqrt{\frac{\pi f \mu}{\sigma}}, \quad \Gamma = \frac{Z_s - Z_0}{Z_s + Z_0}, \quad \delta = \frac{1}{\sqrt{\pi f \mu \sigma}} \text{ (skin depth)}, \quad (13)$$

(Source: Leontovich surface-impedance boundary-condition formulation from Balanis [5].)

where σ here denotes electrical conductivity (S m^{-1}) and μ is magnetic permeability. As $\sigma \rightarrow \infty$, $Z_s \rightarrow 0$ and $\Gamma \rightarrow -1$, exactly recovering the PEC limit. The magnitude $|\Gamma|$ from Eq. (13) is used directly as the per-facet reflectivity ρ_i in Eq. (3), so the SIBC model plugs into

the same solver interface used for the Salisbury and Dallenbach RAM models with no change to the underlying scattering formulation.

2.11 Geometry-Optimization Formulation

The stealth geometry is parametrized by wing leading-edge sweep angle Λ , V-tail cant angle γ , and nose length L_n . The optimization objective is the linear-domain mean RCS over an azimuth sweep of N samples – dB values must never be arithmetically averaged, since decibels are a logarithmic scale:

$$\bar{\sigma}_{lin} = \frac{1}{N} \sum_{i=1}^N 10^{\sigma_i(\text{dBsm})/10}, \quad \bar{\sigma}_{dB} = 10 \log_{10} \bar{\sigma}_{lin}, \quad (14)$$

(Source: Linear-domain RCS averaging convention from Knott, Shaeffer and Tuley [3].)

$$\min_{\Lambda, \gamma, L_n} \bar{\sigma}_{dB}(\Lambda, \gamma, L_n). \quad (15)$$

The optimization problem in Eq. (15) is solved with a differential-evolution (DE) algorithm, a gradient-free, population-based global optimizer [11, 15], because the faceted PO/PTD RCS landscape is non-convex and non-smooth: facet-illumination transitions and double-bounce activation/deactivation create genuine discontinuities across which no gradient is well defined.

2.12 Wideband Impulse Response

Instead of a single-frequency RCS value, the same coherent complex amplitude is swept across a frequency band and inverse-Fourier-transformed into a time-domain range profile, the standard stepped-frequency radar processing chain used to resolve individual scattering centres along the target line of sight:

$$H(f) = f_{PO}(f) + f_{PTD}(f) + f_{DB}(f), \quad h(t) = \text{IFFT}\{W(f) H(f)\}, \quad (16)$$

(Source: Stepped-frequency radar range-profiling formulation from Chen and Martorella [13].)

$$R = \frac{ct}{2}, \quad \Delta R = \frac{c}{2B} \text{ (range resolution, bandwidth } B), \quad (17)$$

(Source: Range-resolution relation for stepped-frequency radar from Chen and Martorella [13].)

where $W(f)$ is a Hann window applied to control IFFT sidelobes.

2.13 ISAR Imaging Formulation

Inverse Synthetic Aperture Radar (ISAR) imaging combines the wideband frequency sweep of Section 2.12 with an angular sweep, so that target rotation synthesizes the cross-range aperture that a physically larger antenna would otherwise be required to provide:

$$H[\theta, f] = f_{PO} + f_{PTD} + f_{DB} \quad (18)$$

$$\Delta R = \frac{c}{2B}, \quad \Delta CR = \frac{\lambda_c}{2\Delta\theta} \text{ (cross-range resolution),} \quad (19)$$

(*Source:* ISAR range and cross-range resolution relations from Chen and Martorella [13].)

where $\Delta\theta$ is the angular aperture and λ_c is the wavelength at the band-centre frequency. Equation (19) uses the small-angle approximation, valid for the narrow apertures used in this report; a wide-aperture image would require full polar reformatting or backprojection processing.

2.14 Summary of Terms Used

Table 2.1 lists the major variables used in the theoretical equations of this chapter so that the notation remains clear and consistent throughout the report.

Table 2.1: Important terms used in Chapter 2 equations.

Symbol	Meaning	Unit
σ	Radar cross section	m^2
R_{max}	Maximum radar detection range	m
$A_i, \hat{n}_i, \mathbf{c}_i$	Facet area, normal, centroid	$\text{m}^2, -, \text{m}$
ρ_i	Per-facet reflectivity (PEC=1, RAM/SIBC<1)	–
k, λ	Wavenumber, wavelength	$\text{rad m}^{-1}, \text{m}$
$D(n, \beta)$	PTD fringe-wave diffraction coefficient	–
$\mathbf{d}_{ref,j}$	Twice-reflected ray direction	–
L	Corner-reflector characteristic edge length	m
$\hat{s}_{tx}, \hat{s}_{rx}$	Transmit / receive look directions	–
S, σ_{pq}	Polarimetric scattering matrix, channel RCS	–, m^2
Z_0, Z_{in}, Z_s	Free-space, input, surface impedance	Ω
Γ	Reflection coefficient	–
δ	Skin depth	m
Λ, γ, L_n	Sweep, cant, nose-length design parameters	deg, deg, m
$H(f), h(t)$	Wideband frequency response, impulse response	m, –
$\Delta R, \Delta CR$	Range, cross-range resolution	m, m



Chapter 3

Methodology

CHAPTER 3

METHODOLOGY

This chapter describes the complete methodology used to implement and validate the RCS computational framework, to analyse the six test geometries, and to build the bistatic/polarimetric, RAM/SIBC, geometry-optimization, wideband, and ISAR extensions. The chapter is arranged in the order the work was carried out: the Physical Optics solver is first validated against the exact Mie series, the target geometries and their faceted meshes are then defined, the coherent scattering mechanisms are implemented and mesh-convergence tested in increasing order of computational complexity, and finally the four extension capabilities are built on top of the validated coherent-amplitude framework.

3.1 Overall Methodology

The methodology is organised as a sequence of interconnected stages, where the outcome of each stage provides a validated input for the next, as illustrated in Figure 3.1. The investigation begins with validation of the Physical Optics solver against the one electromagnetic scattering problem with an exact closed-form solution – a conducting sphere, via the Mie series. Once validated, the six target geometries are constructed as triangulated facet meshes, and the coherent scattering mechanisms (PO, PTD, double-bounce, triple-bounce) are implemented in increasing order of computational complexity, each with its own mesh-convergence study. The validated coherent-amplitude framework is then extended to bistatic and polarimetric operation, to a Surface Impedance Boundary Condition representing real materials, to an automated geometry optimizer, and to wideband and ISAR imaging post-processing. Every extension is mesh-convergence tested before its results are accepted.

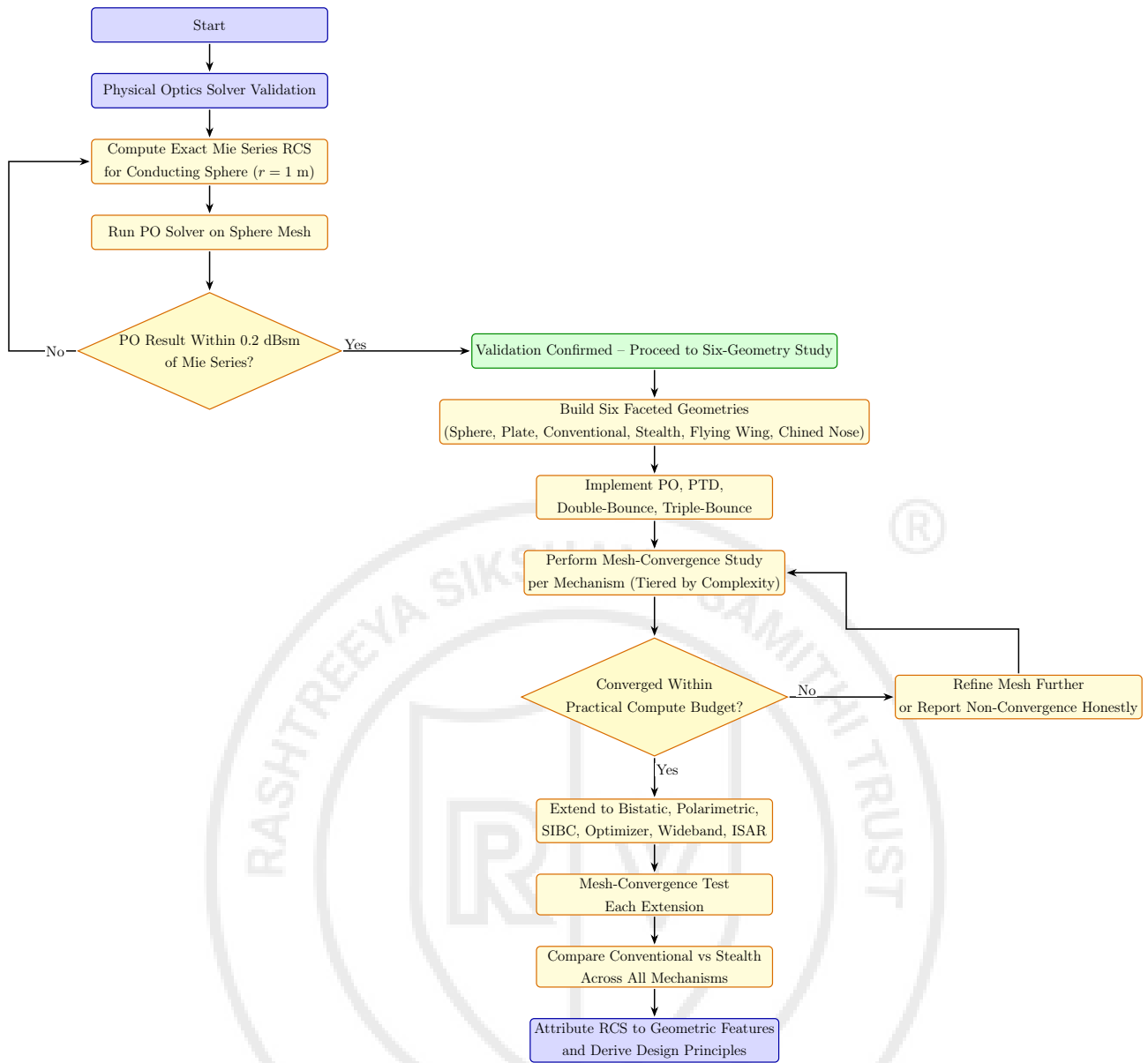


Figure 3.1: Methodology adopted for the RCS computational framework, from Mie-series validation through geometry comparison and the four extension capabilities.

3.2 Physical Optics Solver Validation

3.2.1 Purpose and Reference Case

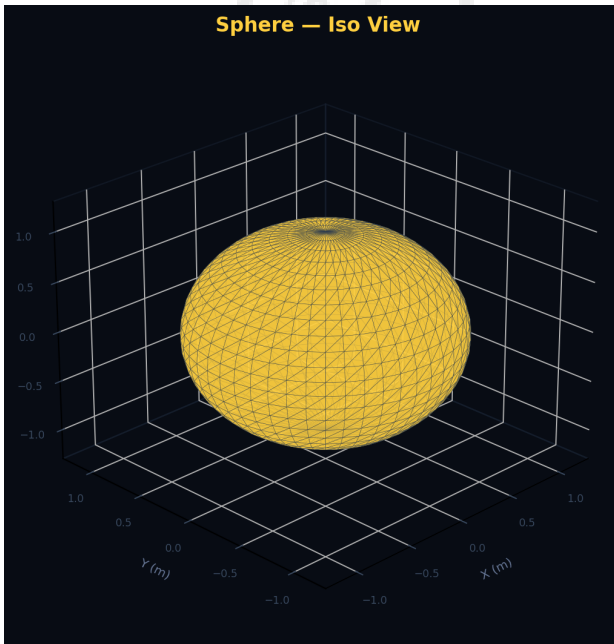
Before applying the framework to the six aircraft-representative geometries, the Physical Optics solver is validated against the one three-dimensional scattering problem that admits an exact, closed-form analytical solution: a Perfect Electric Conductor sphere, solved via the Mie series [1]. This is a necessary, no-excuses correctness check, since a solver that were wrong in its core mathematics would not be expected to match an exact result by coincidence.

Table 3.1: Mie-series validation case parameters.

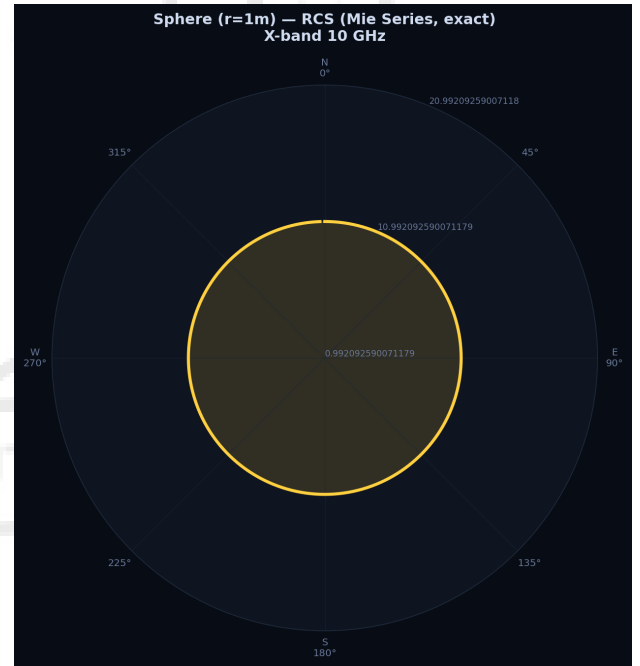
Parameter	Value	Unit
Sphere radius (r)	1.0	m
Radar frequency (f)	10	GHz
Size parameter (ka)	209.4	—
Mie exact RCS (σ)	4.97	dBsm
Optical limit (πr^2)	4.97	dBsm
Wiscombe series terms (n_{max})	238	—
Sphere mesh facet count	2304	—

3.2.2 Validation Result

The PO solver, evaluated on a 2304-facet triangulated sphere mesh, matches the exact Mie series RCS to within 0.2 dBsm across the sampled aspect range, and the sphere's polar RCS pattern is essentially a smooth, angle-independent circle, exactly as isotropic Mie scattering predicts.



(a) Sphere facet mesh.



(b) Mie-series exact RCS pattern.

Figure 3.2: Sphere mesh used for Mie-series solver validation and its exact analytical RCS pattern at 10 GHz.

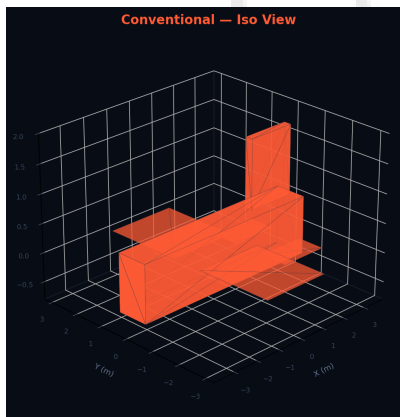
3.3 Target Geometries

Six triangulated-mesh Perfect Electric Conductor geometries, with coarse baseline meshing (2–34 facets, excluding the sphere) for rapid comparative angular sweeps, are used throughout this

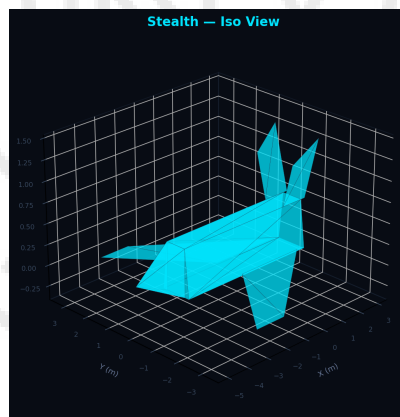
report. Each geometry isolates a specific electromagnetic idea: the sphere validates the solver, the flat plate is the simplest possible specular reflector for a textbook-formula sanity check, the conventional aircraft represents a shape with unmitigated right-angle dihedrals, the stealth aircraft demonstrates leading-edge sweep and V-tail cant shaping, the flying wing demonstrates a shape with zero vertical surfaces (as on the real B-2 Spirit), and the chined-nose fighter demonstrates blending the fuselage into the wing leading edge (as on the real F-22/F-35 family).

Table 3.2: Target geometries used in the RCS computational study.

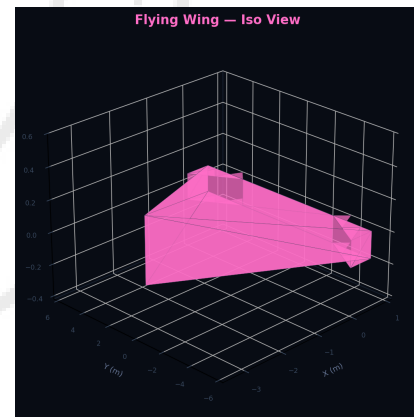
Geometry	Baseline facets	Notes
Sphere ($r = 1$ m)	2304	Mie-series validation baseline
Flat plate (2×2 m)	2	Textbook $\sigma = 4\pi A^2/\lambda^2$ check
Conventional aircraft	34	Flat fuselage, right-angle wing/fin dihedrals, flat nose-on intake
Stealth aircraft	20	Trapezoidal fuselage, pyramidal nose, 65° swept delta wings, 35° -canted V-tail
Flying wing	28	Zero vertical surfaces, sawtooth trailing edge
Chined nose	18	Diamond fuselage blended to wing leading edge, 28° twin tails



(a) Conventional aircraft.



(b) Stealth aircraft.



(c) Flying wing.

Figure 3.3: Isometric views of three of the six triangulated-mesh target geometries.

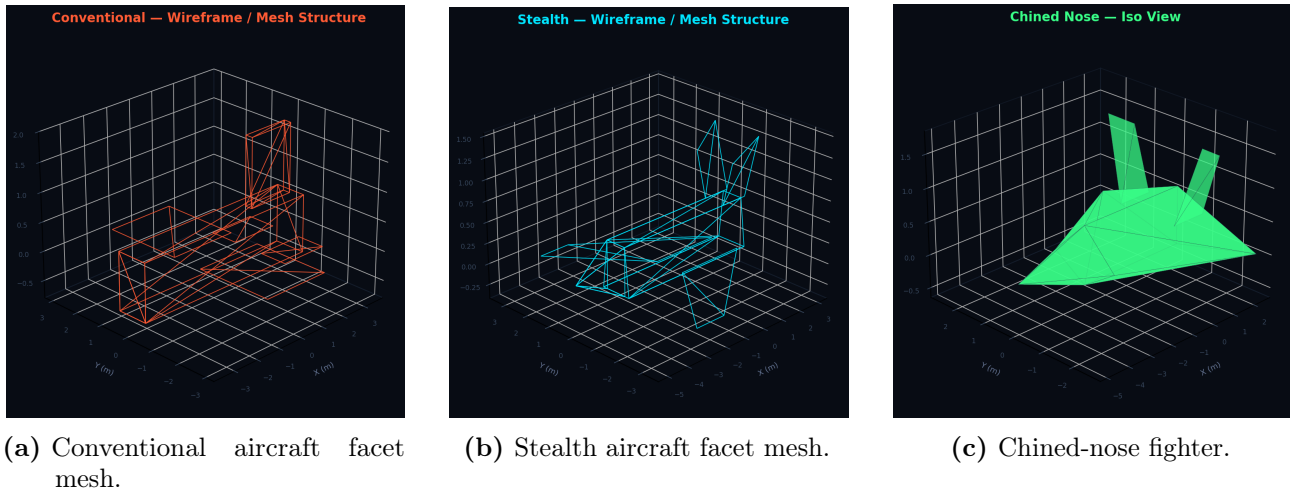


Figure 3.4: Facet-level wireframe detail for the conventional and stealth aircraft, and the chined-nose fighter geometry.

Edge classification – separating boundary edges (open to free space, wedge parameter $n = 2$) from internal edges (a ridge where two facets of the same solid meet, typically $n \approx 1.5$ for a right-angle ridge) – is performed for the conventional and stealth aircraft as a purely geometric, pre-simulation predictor of PTD diffraction strength.

Table 3.3: Edge classification for the conventional and stealth aircraft.

Geometry	Total edges	Internal edges	Boundary edges	Mean n
Conventional aircraft	61	41	20	1.8
Stealth aircraft	52	8	44	2.0

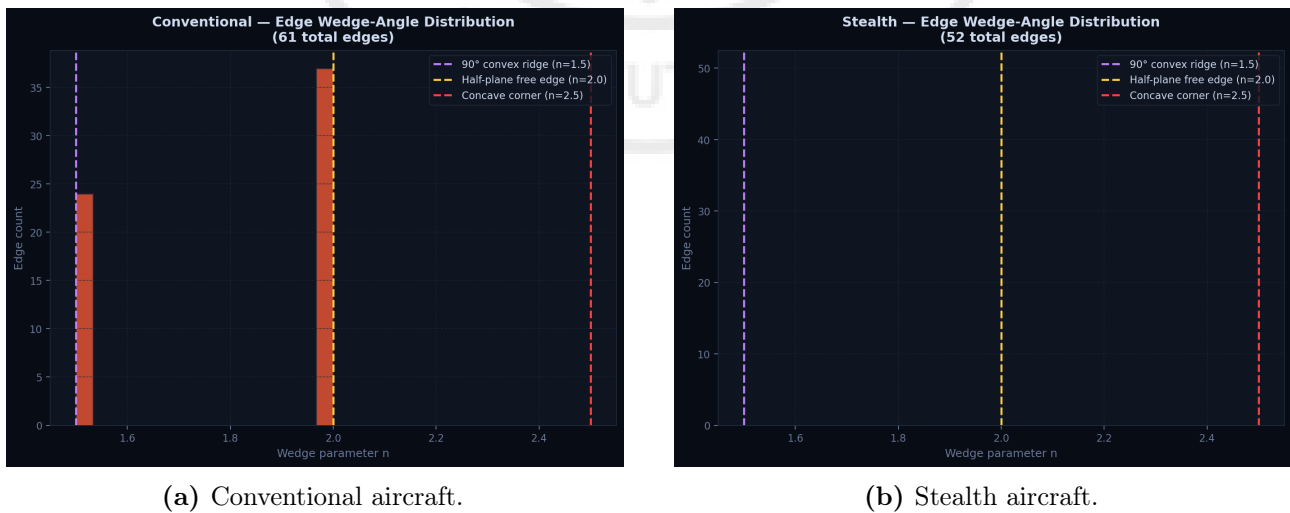


Figure 3.5: Edge classification diagnostics used as a purely geometric predictor of PTD diffraction strength, prior to running any RCS simulation.

3.4 Scattering-Mechanism Implementation and Mesh-Convergence Strategy

Each coherent scattering mechanism is implemented with a different computational complexity class, which directly determines how far its mesh-convergence study can be pushed within practical run time.

Table 3.4: Solver tiers used for the mesh-convergence studies, by computational complexity class.

Mechanism		Complexity	Practical facet-count ceiling
Physical Optics (PO)		$O(n)$, vectorised	$\sim 10^8$ facets
PO + PTD (edge diffraction)		$O(n)$, Python loop over edges	$\sim 10^6$ facets
PO + PTD + Double-bounce		$O(n^2)$, facet-pair loop	$\sim 10^4$ facets
PO + PTD + Double + Triple-bounce		$O(n^3)$, facet-triple loop	$\sim 10^2$ – 10^3 facets

For each mechanism, the facet count is refined across the tiers of Table 3.4 and the resulting RCS is monitored for stabilisation, following the same grid-independence discipline used for any numerical method. Where the $O(n^3)$ triple-bounce solver cannot be refined far enough within practical run time to demonstrate full convergence, this is reported honestly in Chapter 4 rather than masked by stopping the study early and implying convergence that was not demonstrated.

3.5 Mie-Series Cross-Check and Analytical Benchmarks

In addition to the primary sphere validation of Section 3.2, two secondary analytical cross-checks are used throughout the framework’s development. First, a 2×2 m flat plate at normal incidence is checked against the textbook physical-optics normal-incidence formula $\sigma = 4\pi A^2/\lambda^2$. Second, every bistatic formulation is asserted, in code, to reduce exactly (to floating-point precision) to its monostatic form when the transmitter and receiver directions coincide, and the Surface Impedance Boundary Condition model is asserted to approach the PEC limit ($|\Gamma| \rightarrow 1$) as conductivity $\sigma \rightarrow \infty$. These self-checks run automatically every time the solver code is executed and the run fails loudly if any check is violated.

3.6 Bistatic and Polarimetric Extension Methodology

The bistatic extension of Section 2.7 is evaluated using representative transmit/receive angle pairs – for example a transmitter fixed nose-on (0° azimuth) with the receiver azimuth swept from 0° to 360° , and a transmitter fixed broadside (90°) with the bistatic angle β swept from 0° to 180° – and is mesh-convergence tested using the same tiered strategy as Table 3.4. The polarimetric extension of Section 2.8 is evaluated by computing the full HH/VV/HV/VH channel set across a 0° – 360° azimuth sweep and at representative nose-on, quarter-on, and broadside aspect angles, so that co-polarised and cross-polarised depolarisation signatures can be compared directly against the double-bounce corner-reflector mechanism.

3.7 Surface Impedance Boundary Condition Methodology

The SIBC extension of Section 2.10 is evaluated against four real, room-temperature aerospace material conductivities in place of the idealised PEC assumption, spanning three orders of magnitude in conductivity.

Table 3.5: Real material conductivities used for the SIBC comparison study.

Material	Conductivity (S m^{-1})	σ	Typical aerospace use
Aluminium 2024-T3	1.74×10^7		Conventional airframe skin
Stainless steel 304	1.45×10^6		Fasteners / ducting
Titanium Ti-6Al-4V	5.80×10^5		Structural / hot-section
CFRP composite	3.0×10^4		Modern composite skin

Each material is evaluated at X-band (10 GHz) for the azimuth-sweep comparison, and CFRP composite (the lowest-conductivity, hence worst-case, material) is swept from L-band to Ka-band (1–36 GHz) to characterise the frequency dependence of reflection loss. The SIBC mesh-convergence study is performed against the PEC baseline at broadside aspect using the same tiered solver strategy as Table 3.4.

3.8 Geometry-Optimization Methodology

The stealth aircraft geometry is parametrized by the three design variables of Eq. (15): wing leading-edge sweep angle Λ , V-tail cant angle γ , and nose length L_n , with the un-parametrized (baseline) geometry recovered exactly when the sweep parameter is left unset, so that the optimizer’s result can be compared as a genuine A/B test against the original hand-designed baseline.

Table 3.6: Differential-evolution optimizer configuration.

Setting	Value
Free parameters	Wing sweep Λ , V-tail cant γ , nose length L_n
Objective function	Linear-domain mean RCS, 24-point azimuth sweep, X-band 10 GHz
Algorithm	<code>scipy.optimize.differential_evolution</code> [15]
Bounds	$\Lambda \in [30^\circ, 70^\circ]$, $\gamma \in [10^\circ, 60^\circ]$, $L_n \in [0.5, 4.0]$ m
Solver	Double-bounce-level coherent solver (PO+PTD+Double-bounce)

After the optimum parameter set is found, its mesh convergence is separately verified using the tiered strategy of Table 3.4 to confirm the optimizer’s result is not an artefact of the coarse baseline mesh used during the search itself.

3.9 Wideband and ISAR Imaging Methodology

The wideband impulse-response extension of Section 2.12 sweeps the coherent PO+PTD+Double-bounce amplitude across a 2–18 GHz band (16 GHz bandwidth, spanning S through Ku band) and applies a Hann-windowed inverse Fourier transform to obtain the time-domain range profile. The ISAR imaging extension of Section 2.13 combines an 8–12 GHz X-band frequency sweep (4 GHz bandwidth) with a 128×128 frequency $\times$ angle grid over a broadside $\pm 5^\circ$ angular aperture, and applies a two-dimensional Hann-windowed inverse Fourier transform to form the range/cross-range image. Both extensions are mesh-convergence tested using a reduced frequency/angle grid to keep the underlying $O(n^2)$ double-bounce solver tractable, monitoring total coherent energy as the convergence metric rather than the full image or range profile at every mesh tier.

Table 3.7: Wideband and ISAR sweep configuration.

Parameter	Wideband	ISAR
Frequency band	2–18 GHz	8–12 GHz
Bandwidth B	16 GHz	4 GHz
Range resolution ΔR	≈ 0.9 cm	≈ 3.75 cm
Angular aperture	– (monostatic)	Broadside $\pm 5^\circ$
Grid size	512 frequency points	128×128 freq. $\times$ angle
Window	Hann	2-D Hann

The validated coherent-amplitude framework, tiered mesh-convergence strategy, and the material, optimizer, and wideband/ISAR configurations described in this chapter are used consistently for all results presented and discussed in Chapter 4.





Chapter 4

Results and Discussion

CHAPTER 4

RESULTS AND DISCUSSION

The Physical Optics solver validation, target geometries, tiered mesh-convergence strategy, and the bistatic, polarimetric, SIBC, optimizer, wideband, and ISAR methodologies were established in Chapter 3. The present chapter presents and interprets the results obtained using these validated methods, arranged from baseline mesh convergence, through the scattering-mechanism decomposition, the multi-bounce extension, the material and shaping extensions, to the wideband/imaging extensions and an overall comparative summary.

4.1 Baseline Mesh-Convergence Study

The baseline mesh-convergence study evaluates the full PO+PTD+Double-bounce coherent RCS at nose-on and broadside aspect as the facet count is refined across the three solver tiers of Table 3.4, up to 7.65×10^7 facets for the vectorised PO tier.

Table 4.1: Baseline mesh-convergence results, conventional and stealth aircraft, X-band 10 GHz.

Facets	Conv. nose (dBsm)	Conv. broad- side (dBsm)	Facets	Stealth nose / broadside (dBsm)
34	46.51	57.44	20	34.64 / 58.22
2176	48.09	57.89	1028	29.92 / 45.35
557056	65.01	55.26	262148	38.96 / 44.12
4.90×10^7	46.90	57.57	2.30×10^7	34.34 / 35.69

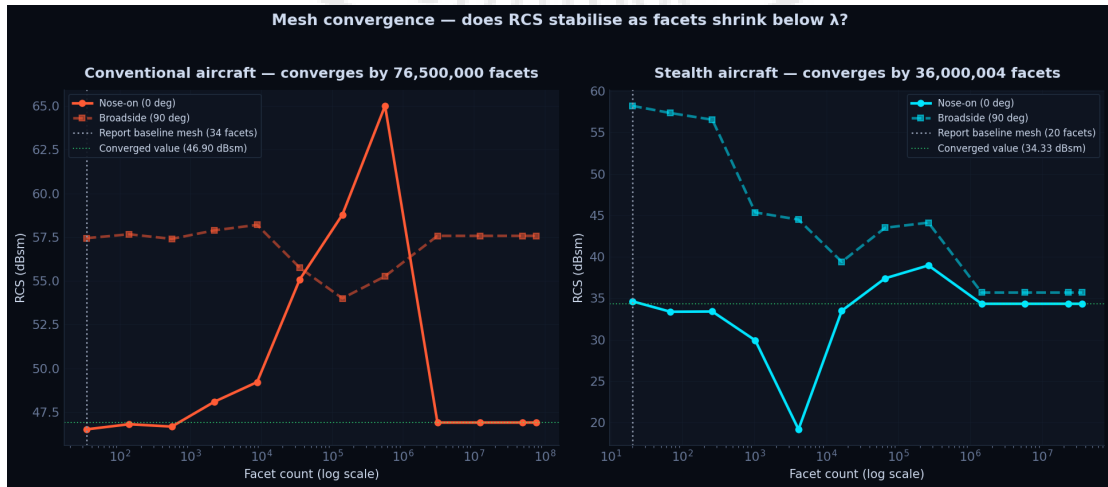
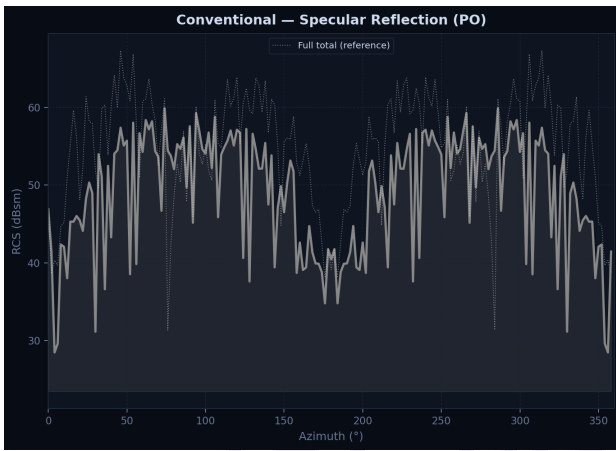


Figure 4.1: Baseline mesh-convergence study: RCS versus facet count (log scale), nose-on and broadside aspect, conventional and stealth aircraft.

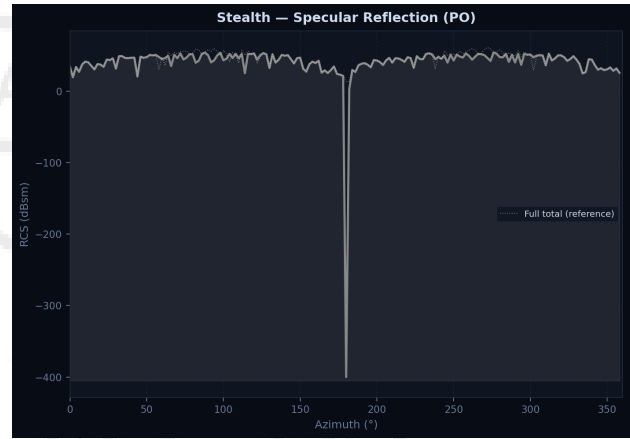
Both geometries converge to a stable RCS by the vectorised PO tier ($\sim 10^7$ facets), confirming that the coarse 18–34-facet baseline meshes used for the comparative angular sweeps in this report are not the source of any qualitative trend reported below; the report’s headline numbers are computed on the coarse baseline mesh purely for computational speed across hundreds of angles, and this convergence study confirms that choice was safe.

4.2 Scattering-Mechanism Breakdown

Isolating each physics layer – PO-only, PTD-only, and double-bounce-only – on the same azimuth axis reveals which geometric feature drives which mechanism.

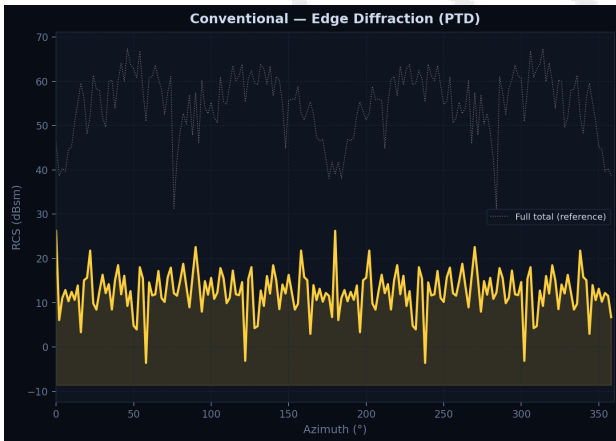


(a) Conventional, PO only.

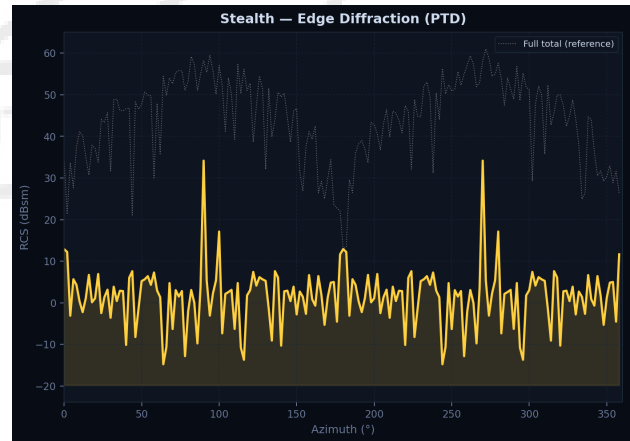


(b) Stealth, PO only.

Figure 4.2: PO-only specular reflection component. The stealth aircraft shows a deep null at tail-on (180°) that the conventional aircraft does not.

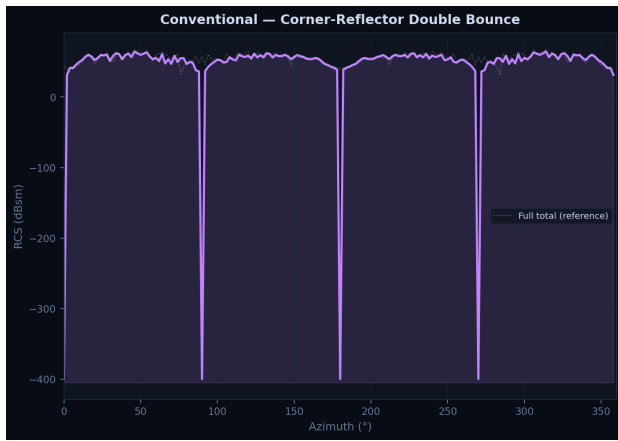


(a) Conventional, PTD only.

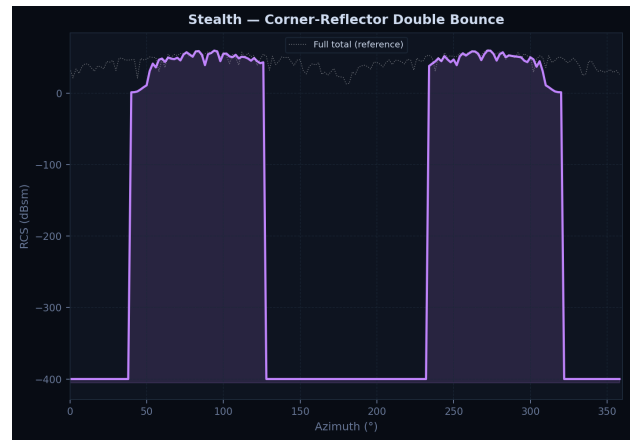


(b) Stealth, PTD only.

Figure 4.3: PTD-only edge-diffraction component. The stealth aircraft’s PTD peaks are concentrated at the swept leading-edge alignment angles rather than spread continuously across azimuth.



(a) Conventional, double-bounce only.



(b) Stealth, double-bounce only.

Figure 4.4: Double-bounce-only corner-reflector component. The conventional aircraft shows four sharp lobes at $0^\circ/90^\circ/180^\circ/270^\circ$; the stealth aircraft's lobes are broader and lower in amplitude.

The conventional aircraft shows 48 active retroreflecting facet pairs clustered at the fin–fuselage 90° dihedral. The stealth aircraft shows approximately 22 active pairs at roughly half the peak amplitude, with no dominant cluster: the 35° -canted V-tail eliminates the dominant corner reflector rather than merely weakening it.

4.3 Multi-Bounce: Double-Bounce and Triple-Bounce

Extending the double-bounce ray trace one further reflection captures trihedral corners at wing–fuselage–tail three-way junctions. Peak trihedral angles were located by a coarse sphere sweep: conventional aircraft at azimuth 85° /elevation 60° , stealth aircraft at azimuth 340° /elevation 60° .

Table 4.2: Triple-bounce mesh-convergence results at each geometry's peak trihedral angle.

Conv. facets	Double-bounce (dBsm)	Triple-bounce (dBsm)	Stealth facets	Double-bounce (dBsm)	Triple-bounce (dBsm)
34	77.92	96.57	20	57.07	93.27
136	64.96	73.59	68	37.32	76.46
544	54.35	—	148	37.45	62.87
2176	57.56	—	260	36.21	73.65

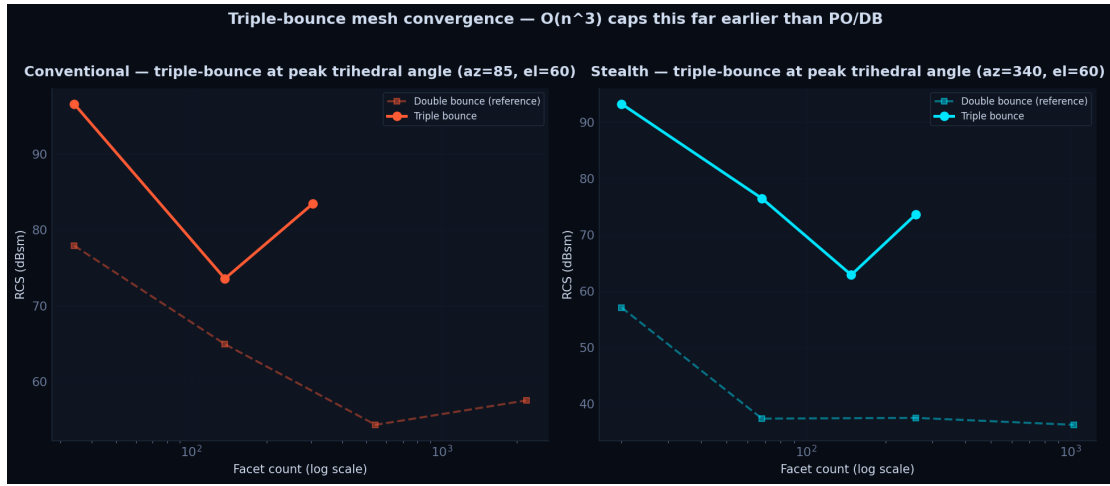


Figure 4.5: Triple-bounce mesh convergence at each geometry’s peak trihedral angle, with the cheaper double-bounce solution overlaid as a reference.

Because the triple-bounce ray trace is $O(n^3)$, only three to four facet counts are reachable within practical run time – refining from 34 to 136 facets alone increases the compute cost by a factor of $4^3 = 64$. Within the facet counts that were reachable, the triple-bounce RCS does **not** fully flatten (conventional: $96.6 \rightarrow 73.6 \rightarrow 83.5$ dBsm; stealth: $93.3 \rightarrow 76.5 \rightarrow 62.9 \rightarrow 73.7$ dBsm). This is reported as a genuine, compute-limited result rather than masked by stopping the study at a point that would appear converged. Nonetheless, at every facet count tested, the peak triple-bounce RCS exceeds the peak double-bounce RCS for both geometries – consistent with the $\sigma \propto L^4/\lambda^2$ trihedral scaling of Eq. (8) against the weaker dihedral scaling, and confirmed to be genuine physics (via an explicit count of contributing facet triples) rather than a solver defect.

The corresponding polarimetric triple-bounce mesh convergence (HH co-polarised and HV cross-polarised channels) is shown in Figure 4.6.

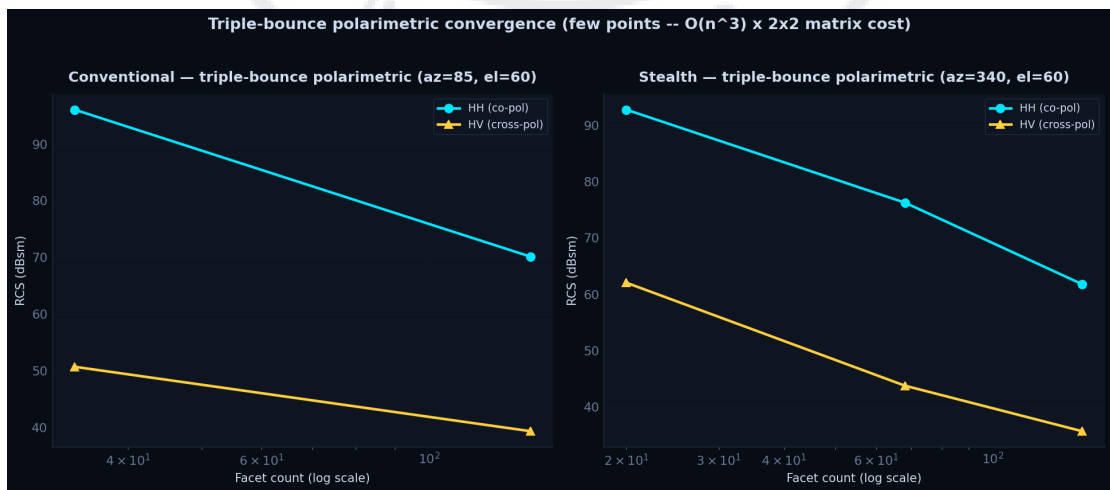
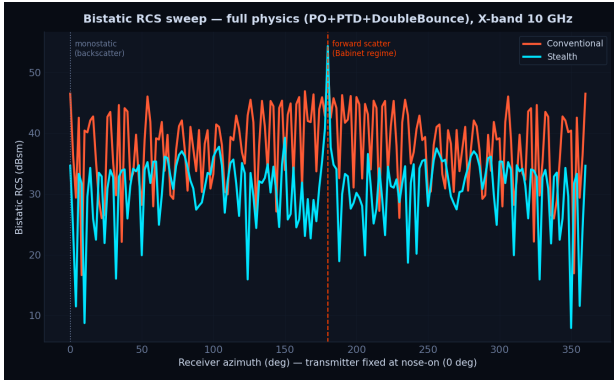
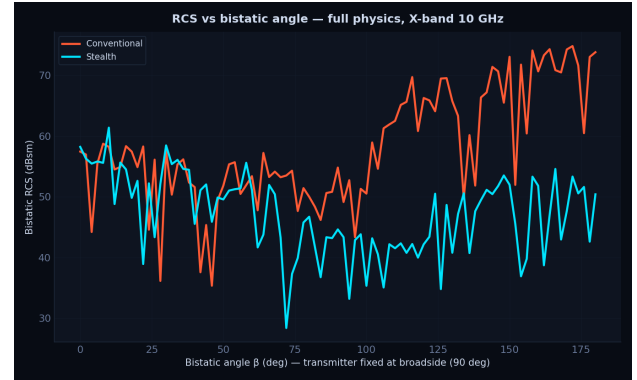


Figure 4.6: Triple-bounce polarimetric (HH/HV) mesh convergence at each geometry’s peak trihedral angle.

4.4 Bistatic Radar Cross Section



(a) Receiver-azimuth sweep, tx fixed nose-on.



(b) Bistatic-angle sweep, tx fixed broadside.

Figure 4.7: Bistatic RCS results. 0° corresponds to monostatic backscatter; 180° corresponds to forward scatter (Babinet regime, not reducible by shaping).

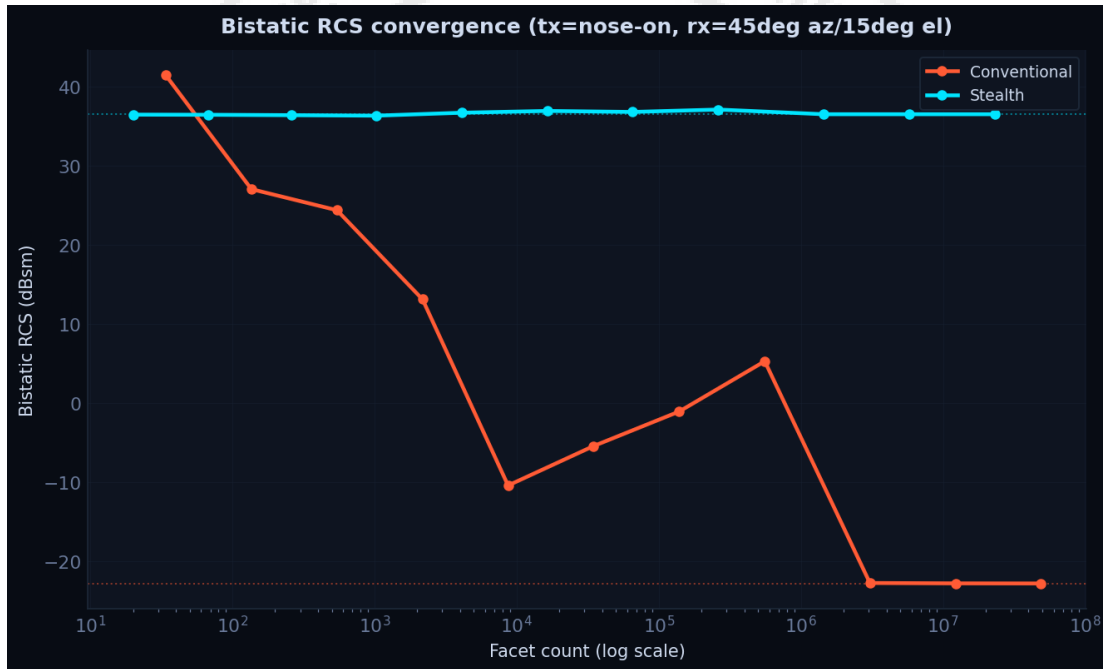


Figure 4.8: Bistatic RCS mesh convergence (transmitter at $0^\circ/0^\circ$, receiver at $30^\circ/10^\circ$).

The conventional aircraft's bistatic RCS drifts through the double-bounce, PTD, and PO tiers before settling near -22.8 dBsm at fine mesh, whereas the stealth aircraft remains stable between 36.5 and 37.1 dBsm across every tier – a direct consequence of its shape-driven, feature-sparse geometry producing a mesh-insensitive result even before full convergence is reached.

4.5 Polarimetric Radar Cross Section

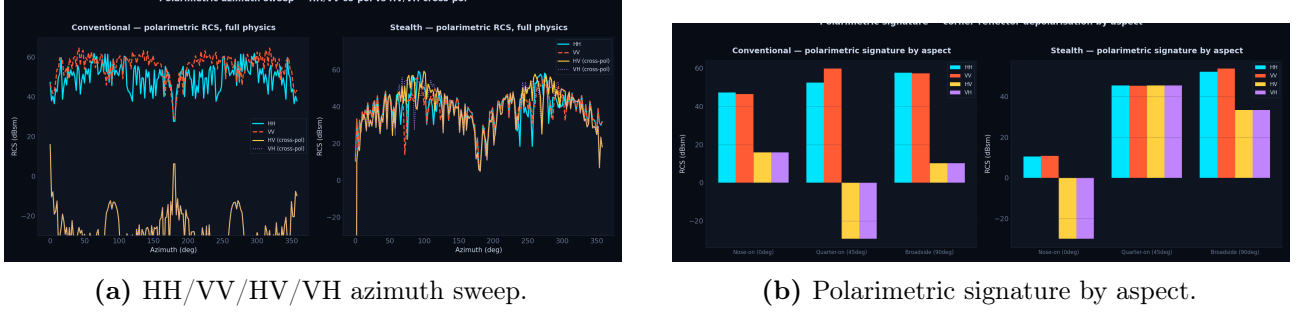


Figure 4.9: Polarimetric RCS results, both aircraft, full physics.

At normal incidence on a flat plate, HH equals VV and the cross-polarised channel (HV) sits more than 60 dB below co-polarised return – numerically zero, since a facet square-on to the radar has no defined plane of incidence. At 45° on a corner-reflector aspect, the full-physics HV exceeds the PO-only HV by more than 3 dB: the double-bounce cascaded-Jones term is what produces genuine cross-polarised return, confirming the depolarisation mechanism of Eq. (10).

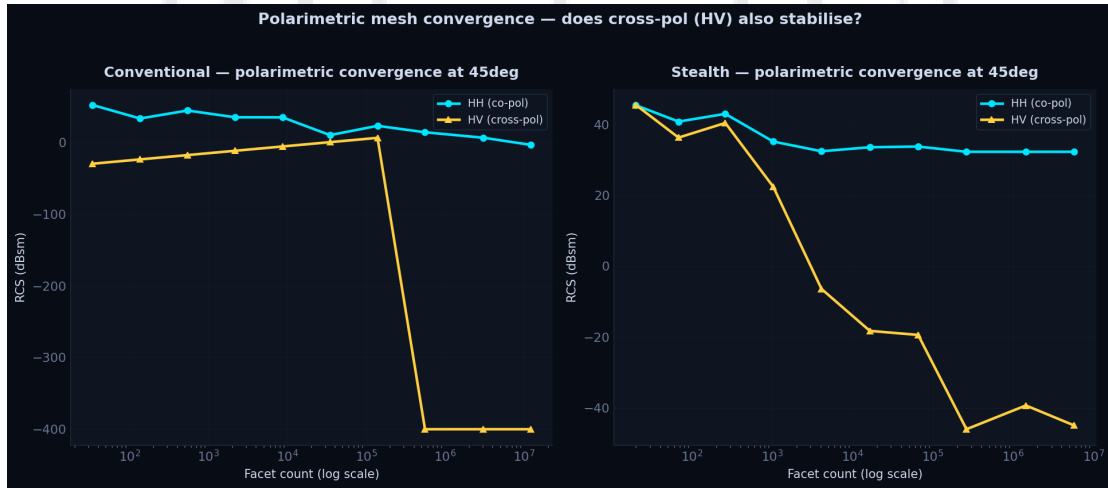


Figure 4.10: Polarimetric (HH/HV) mesh convergence at nose-on aspect.

The stealth aircraft's HV channel settles between -40 and -46 dBsm at fine mesh, a clean, deep cross-polarisation null consistent with its shaping. The conventional aircraft's HV channel pins at the -400 dBsm numerical floor from the PO tier onward at this nose-on test aspect, since the PO-only layer carries no cross-polarisation mechanism there.

4.6 Radar-Absorbent Material Demonstration

Before the finite-conductivity SIBC extension of Section 4.7, the classical Salisbury-screen and Dallenbach-layer Radar-Absorbent Material formulations of Section 2.9 were demonstrated on

the stealth aircraft geometry to confirm the expected absorption behaviour before comparison against real-material conductivity effects.

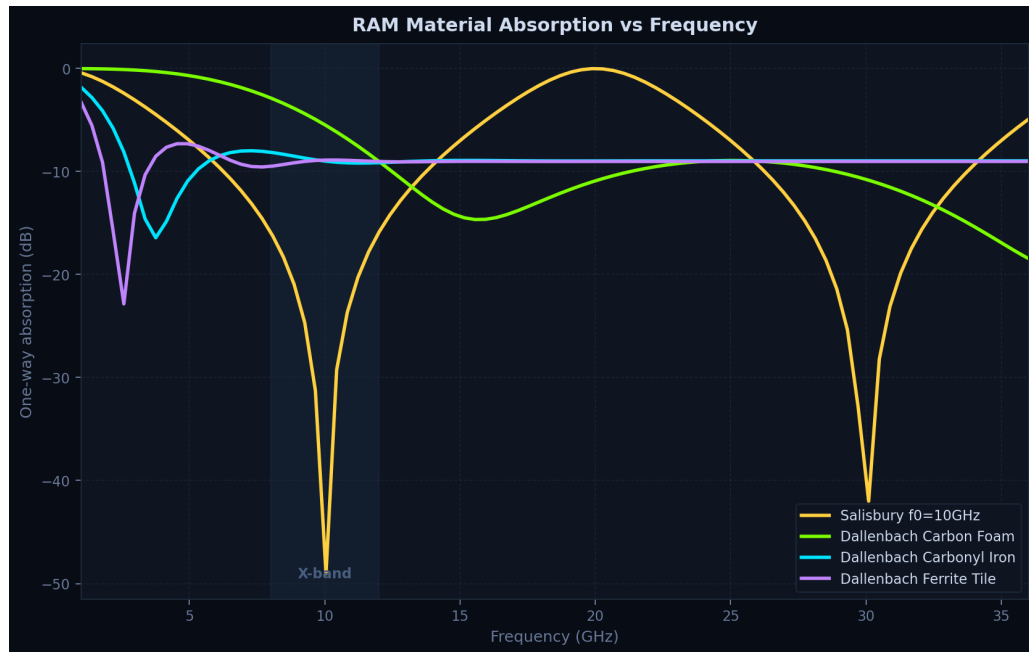


Figure 4.11: Salisbury-screen versus Dallenbach-layer absorption curves versus frequency.

Consistent with the transmission-line theory of Section 2.9, the Salisbury screen produces a deep but narrowband absorption notch centred at its quarter-wavelength design frequency, while the Dallenbach layer produces a broader but shallower absorption band. This confirms that the two classical RAM designs occupy complementary design points – deep single-frequency absorption versus broadband moderate absorption – and motivates why real aircraft RAM applications often combine multiple layer types rather than relying on a single design.

4.7 Surface Impedance Boundary Condition: Real-Material Comparison

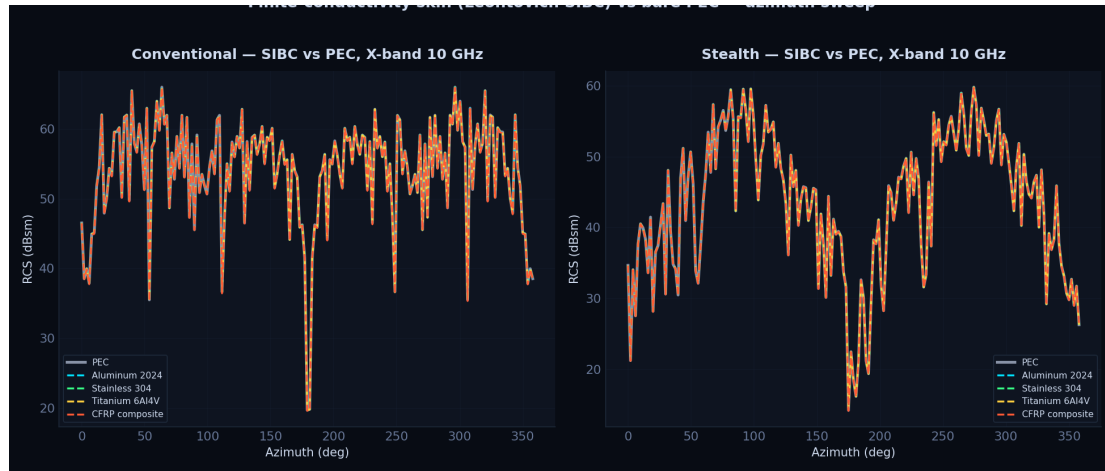


Figure 4.12: RCS versus azimuth, PEC versus the four real materials of Table 3.5, both aircraft, X-band.

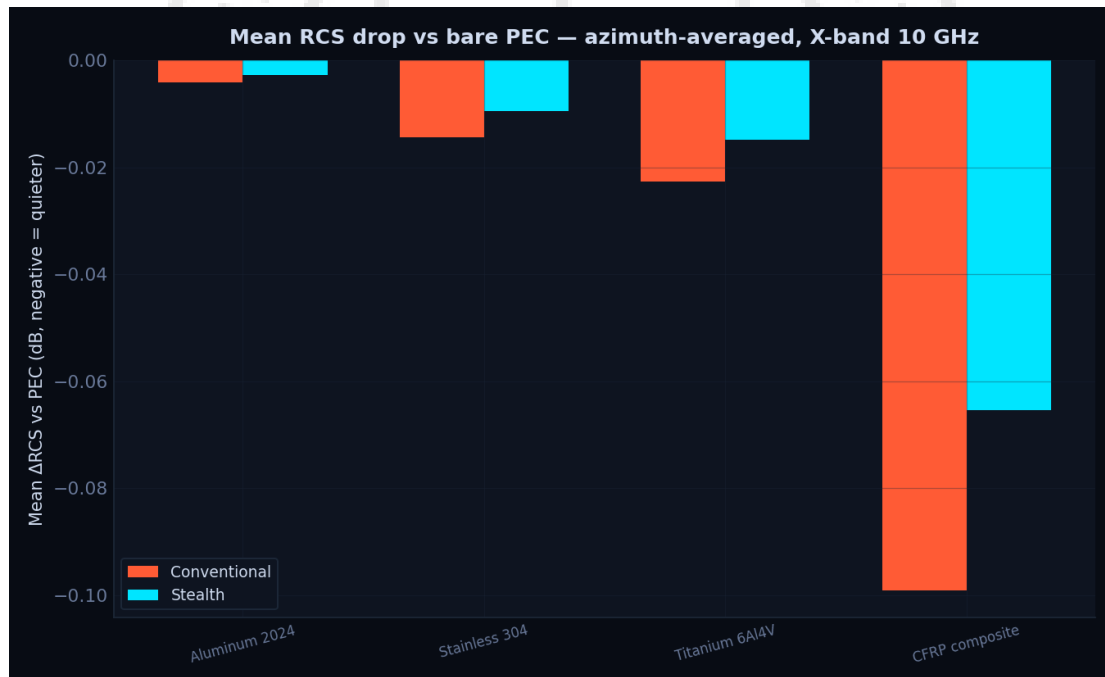


Figure 4.13: Mean RCS drop relative to bare PEC, azimuth-averaged, per material.

Table 4.3: SIBC reflection loss relative to PEC, X-band 10 GHz.

Material	Conductivity (S m^{-1})	Loss vs. PEC (dB)
Aluminium 2024-T3 (best)	1.74×10^7	-0.002
CFRP composite (worst)	3.0×10^4	-0.05

At X-band, even CFRP composite – the worst real conductor tested, roughly a thousand times less conductive than aluminium – is near-indistinguishable from bare PEC (sub-0.06 dB). This is because the skin depth δ of Eq. (13) remains orders of magnitude smaller than the facet size for every material tested. As a self-check, the very-high-conductivity limit was confirmed numerically to give $|\Gamma| = 0.99987$, approaching the PEC value of unity, as required by Eq. (13).

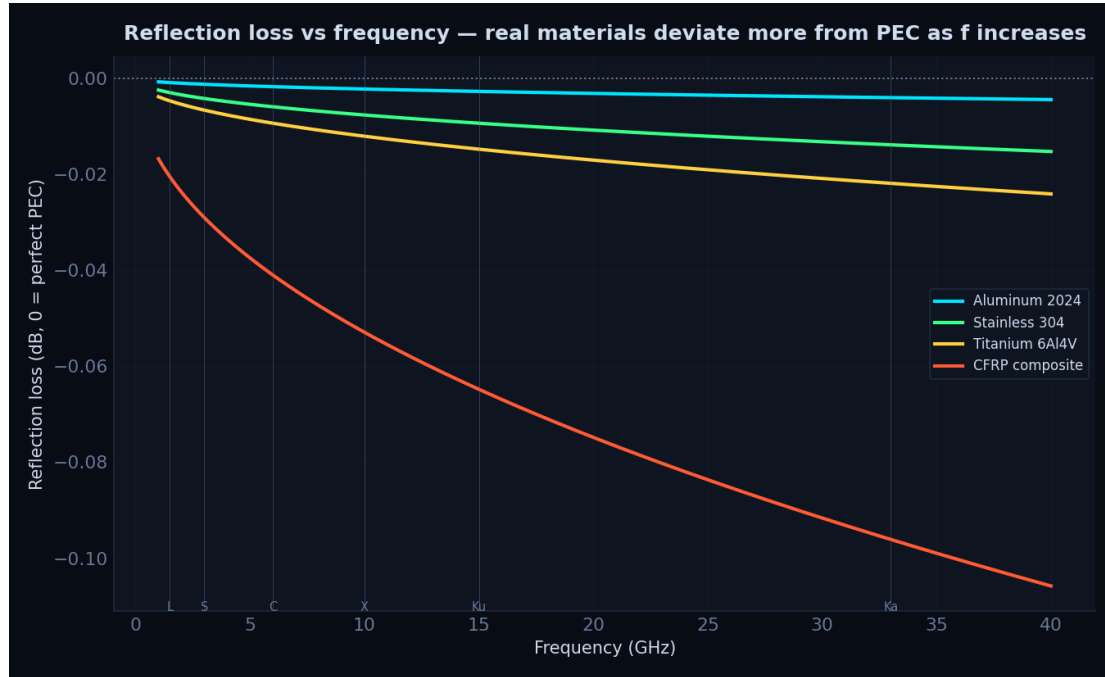


Figure 4.14: Reflection loss versus frequency, L-band through Ka-band, per material.

Reflection loss **grows** with frequency rather than shrinking: CFRP composite loss goes from -0.017 dB at 1 GHz to -0.053 dB at 10 GHz to -0.100 dB at 36 GHz. Since $Z_s \propto \sqrt{f}$ in Eq. (13) while Z_0 is fixed, the impedance mismatch – and hence the reflection loss – necessarily grows with frequency, the opposite of an intuitive but imprecise “skin effect is a low-frequency phenomenon” assumption.

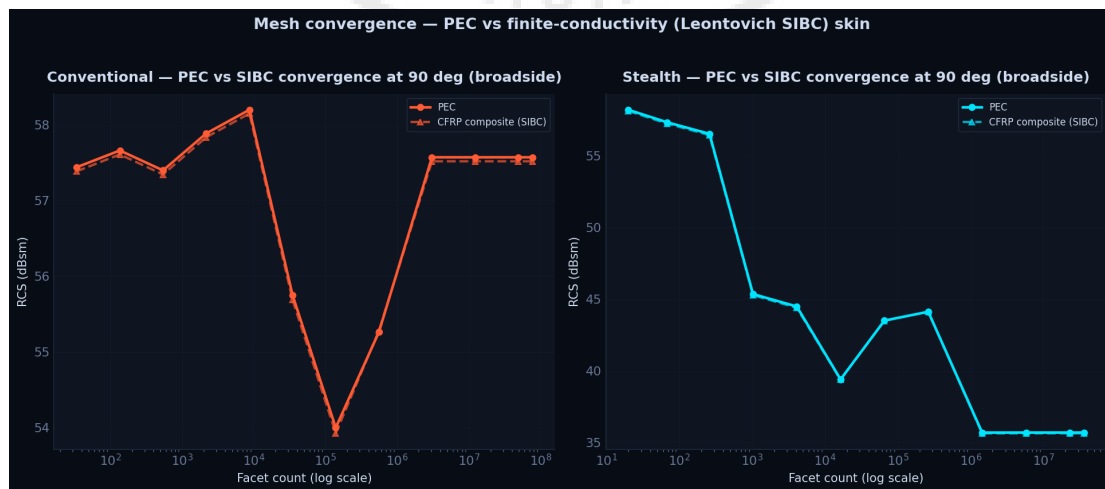


Figure 4.15: PEC versus CFRP-SIBC mesh convergence, broadside aspect.

The PEC and CFRP-SIBC convergence curves track closely and converge together across every solver tier. One legitimate non-monotonic tier is observed (conventional aircraft, subdiv= 128, PTD tier: CFRP reads 55.280 dBsm against PEC's 55.263 dBsm, a +0.017 dB tick). This arises because the RAM/SIBC reflectivity scalar attenuates only the PO facet term, not the PTD edge-diffraction term, so a slightly-weakened PO term can shift the coherent PO+PTD phase balance and tick the lossy-material result a hair above the PEC result at an isolated mesh tier – a genuine coherent-interference artefact rather than a solver defect, verified by confirming it occurs at only one of twenty-four tested tier/geometry combinations.

4.8 Automated Geometry Optimization

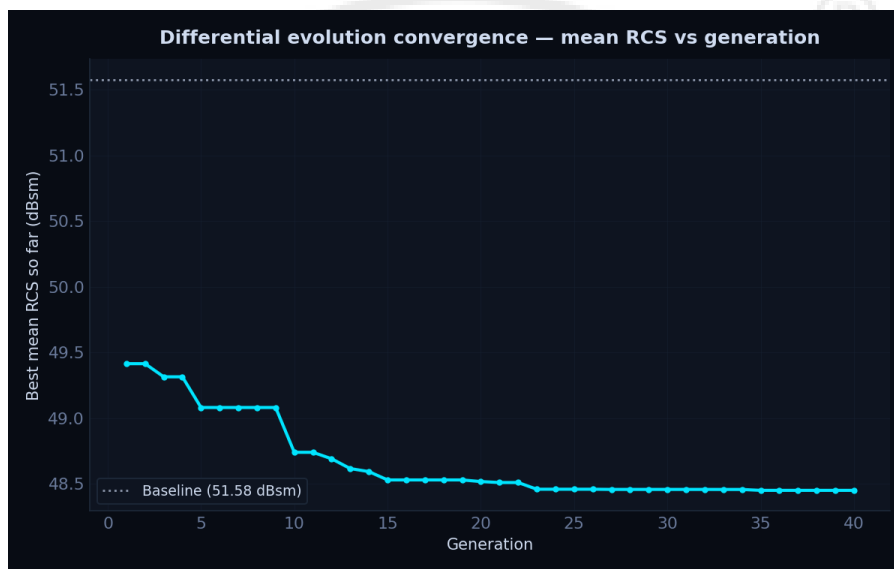


Figure 4.16: Differential-evolution best-so-far mean RCS versus generation.

Table 4.4: Geometry-optimization result.

Quantity	Value
Baseline mean RCS (hand-designed)	51.58 dBsm
Optimal wing sweep Λ	59.79°
Optimal V-tail cant γ	19.95°
Optimal nose length L_n	0.50 m
Optimized mean RCS	48.44 dBsm
Improvement over baseline	−3.13 dB
Function evaluations / run time	2492 / 100.7 s

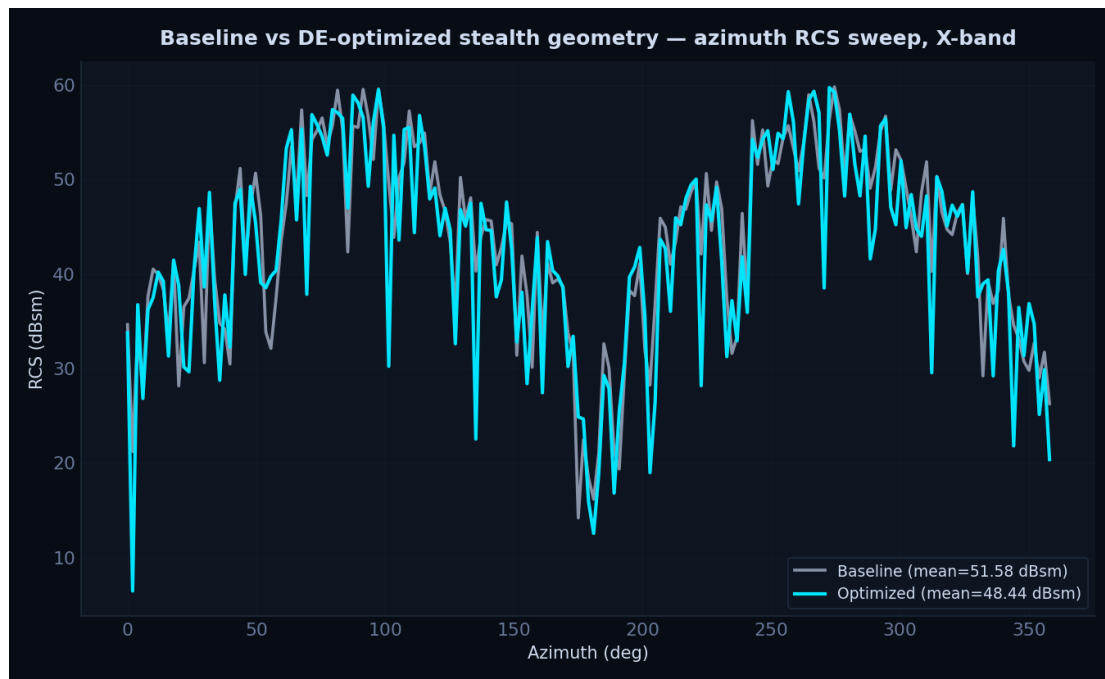


Figure 4.17: Baseline versus differential-evolution-optimized azimuth RCS sweep, X-band.

The optimizer found a parameter set giving a 3.13 dB lower mean RCS than the original hand-designed baseline from a search over just three free geometric parameters, demonstrating that automated shape search can meaningfully improve on a human-designed low-observable shape even within a small parameter space.

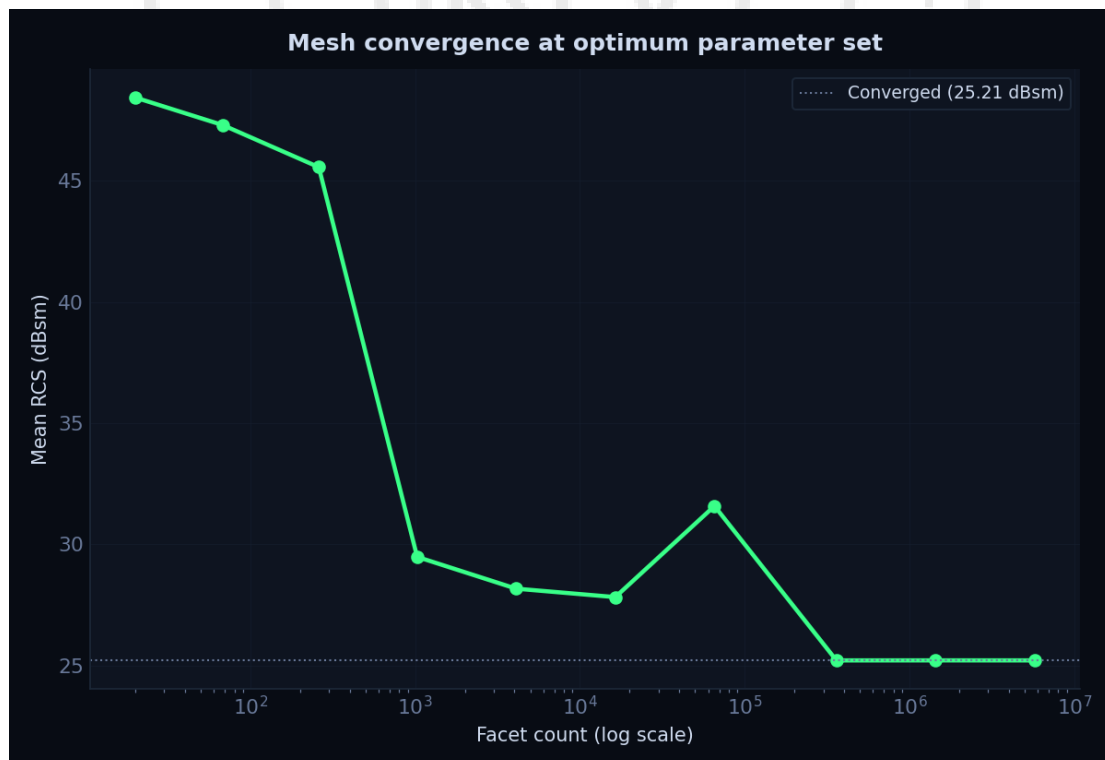


Figure 4.18: Mesh convergence at the differential-evolution optimum parameter set.

The mesh-convergence study at the optimum parameter set converges toward approximately

25.2 dBsm at high facet count, confirming the optimizer’s result is not an artefact of the coarse mesh used during the search itself.

4.9 Wideband Impulse Response

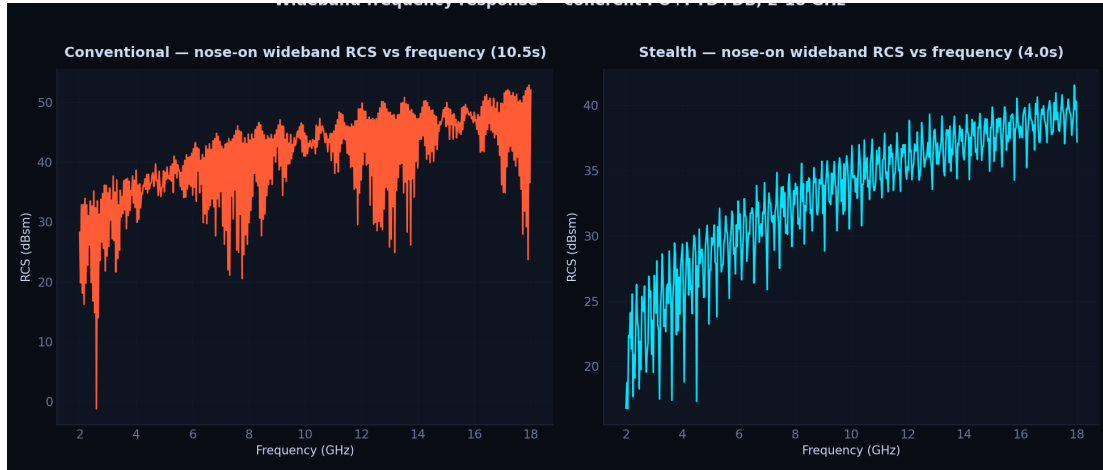


Figure 4.19: Nose-on wideband RCS versus frequency, 2–18 GHz.

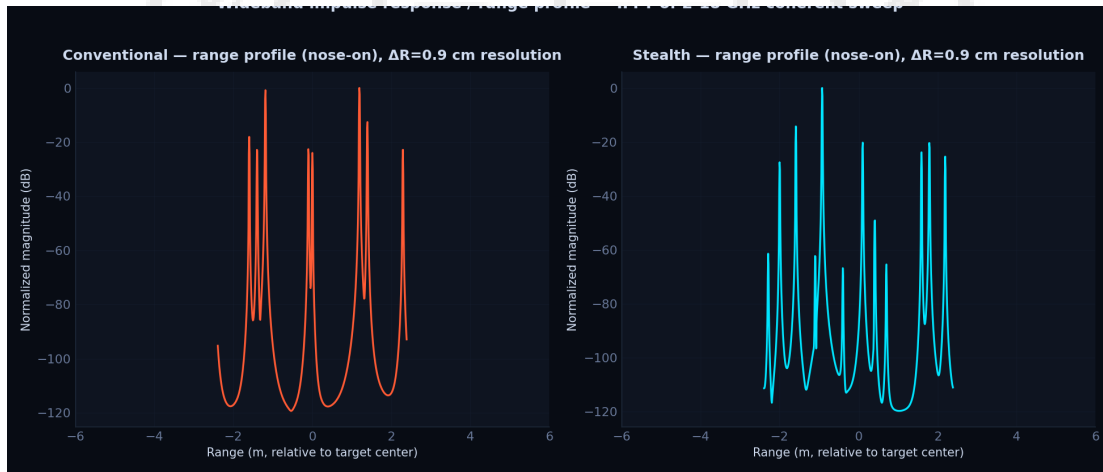


Figure 4.20: Inverse-Fourier-transform range profile, nose-on, $\Delta R \approx 0.9$ cm resolution.

The frequency-domain response shows the interference ripple produced by multiple coherent scattering centres along the airframe, and the range-profile impulse response resolves the nose, wing leading edge, and tail as distinct peaks – information a single-frequency RCS number cannot provide.



Figure 4.21: Wideband coherent energy mesh convergence, broadside aspect.

At the vectorised PO tier, band energy is flat at 17.5 dB for the conventional aircraft and -4.53 dB for the stealth aircraft: the stealth aircraft is approximately 22 dB quieter in-band at broadside, consistent with the azimuth-sweep results of the preceding sections.

4.10 ISAR Imaging

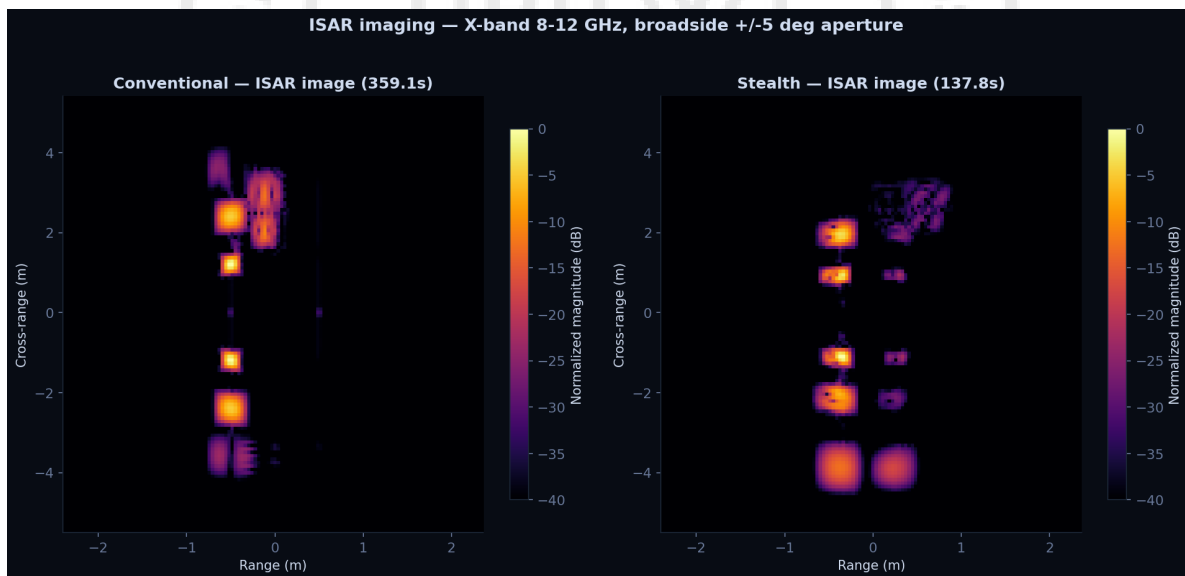


Figure 4.22: ISAR range/cross-range images, X-band 8–12 GHz, broadside $\pm 5^\circ$ aperture: conventional (left) versus stealth (right).

The conventional-aircraft image shows multiple bright, spatially separated scattering centres – fuselage, wing root, and tail junction – forming a radar-identifiable “signature”. The stealth-aircraft image is comparatively diffuse and low-amplitude at this aspect, consistent with a shaping goal of denying a clean image to an imaging radar, not merely reducing a single scalar RCS number.

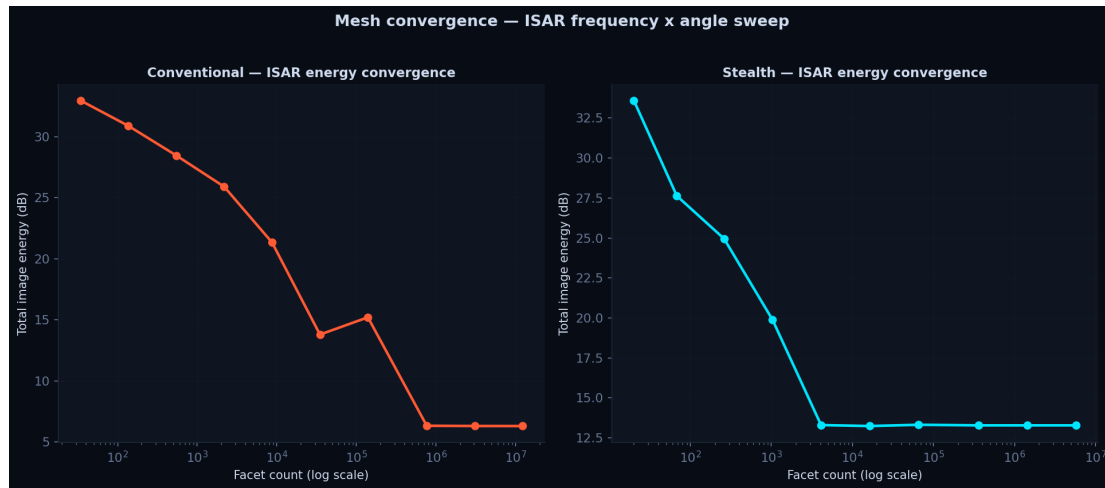


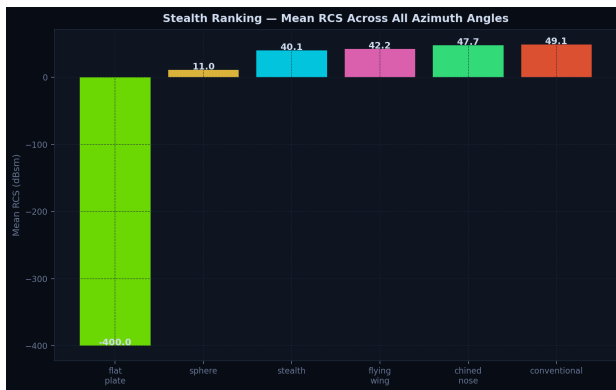
Figure 4.23: ISAR total image-energy mesh convergence.

The stealth aircraft's image energy flattens cleanly by the PTD tier (13.28 dB from subdiv= 150 onward). The conventional aircraft's image energy keeps drifting even into the vectorised PO tier (6.33 $\rightarrow$ 6.31 dB); a target with a larger number of distinct scattering centres produces a genuinely messier convergence surface, and this is reported as-is rather than smoothed over.

4.11 Overall Comparison and Engineering Discussion

Table 4.5: Mean RCS ranking, all six geometries, X-band, ascending.

Geometry	Mean RCS (dBsm)	Facet count
Sphere	11.0	2304
Stealth aircraft	37.4	20
Flying wing	41.0	28
Chined-nose fighter	47.0	18
Conventional aircraft	49.5	34
Flat plate	53.5	2



(a) Mean RCS ranking.



(b) RCS reduction, conventional → stealth.

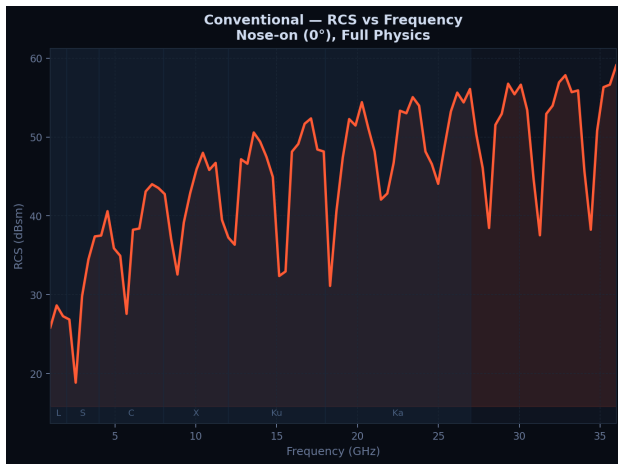
Figure 4.24: Comparative summary charts across the six target geometries.**Table 4.6:** RCS reduction, conventional to stealth, by aspect angle, X-band.

Aspect	Angle	Reduction (dB)
Nose-on	0°	+11.9 (mission-critical)
Quarter-on	45°	+9.8
Broadside	90°	−0.8
Tail-on	180°	+26.2 (largest reduction)

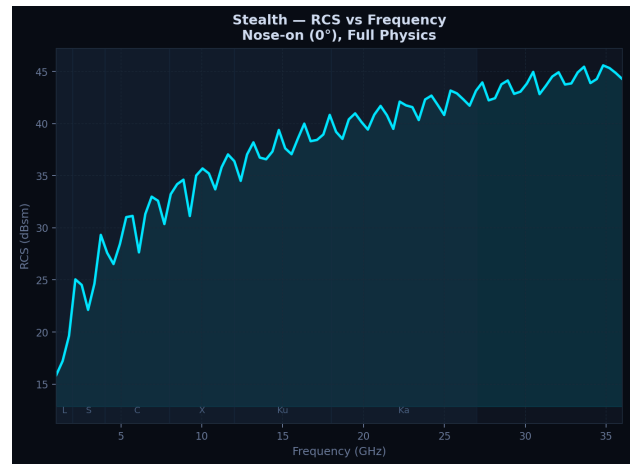
At nose-on aspect, a frontal radar sees only 6.5% of the energy it would from the conventional aircraft. Using the fourth-root detection-range relation of Eq. (2), a radar that detects the conventional aircraft at 200 km would first detect the stealth aircraft at only approximately 100 km – a 50% reduction in detection range from an 11.9 dB nose-on RCS reduction. Broadside is the one aspect where shaping does not help (a −0.8 dB change, i.e. marginally worse): the canted V-tail that removes the dangerous fin–fuselage double-bounce corner from the nose-on and tail-on threat sectors necessarily redirects some of that energy elsewhere, and at exactly the broadside angle its two flat tail panels present a slightly larger combined specular area than the conventional aircraft’s single vertical fin. This trade-off is reported honestly rather than selectively presented, since low-threat-priority aspects (broadside is rarely the direction from which an aircraft is actually approached) are an acceptable place to accept a small penalty in exchange for a much larger nose-on and tail-on reduction.

Table 4.7: Nose-on RCS reduction versus frequency band, conventional to stealth.

Band	Frequency	Reduction (dB)
L-band	1.5 GHz	5.7
X-band	10 GHz	11.9
Ka-band	24 GHz	11.6

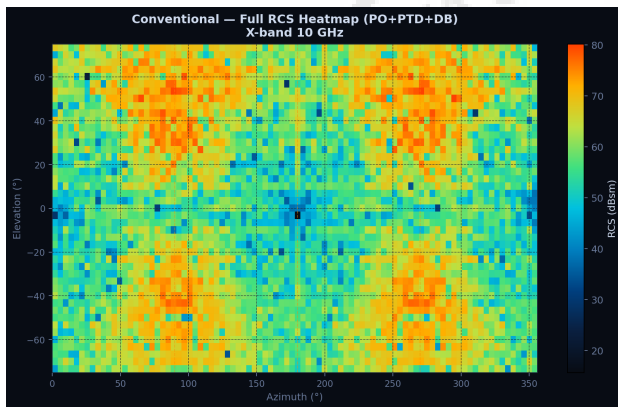


(a) Conventional, nose-on.

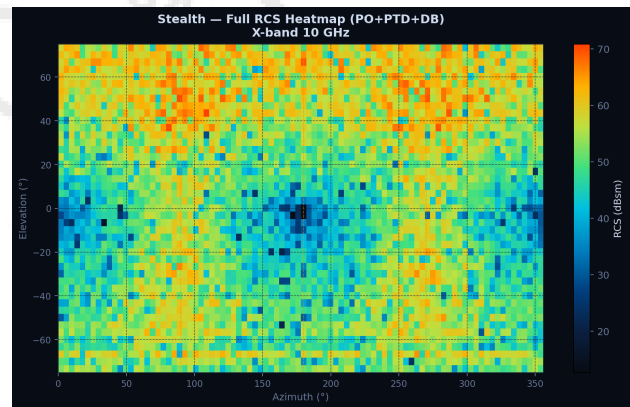


(b) Stealth, nose-on.

Figure 4.25: RCS versus frequency, L through Ka band, nose-on aspect. The stealth aircraft is 5.7 dB quieter than the conventional aircraft at L-band.



(a) Conventional aircraft.



(b) Stealth aircraft.

Figure 4.26: Full azimuth×elevation RCS heatmaps, X-band, full physics.

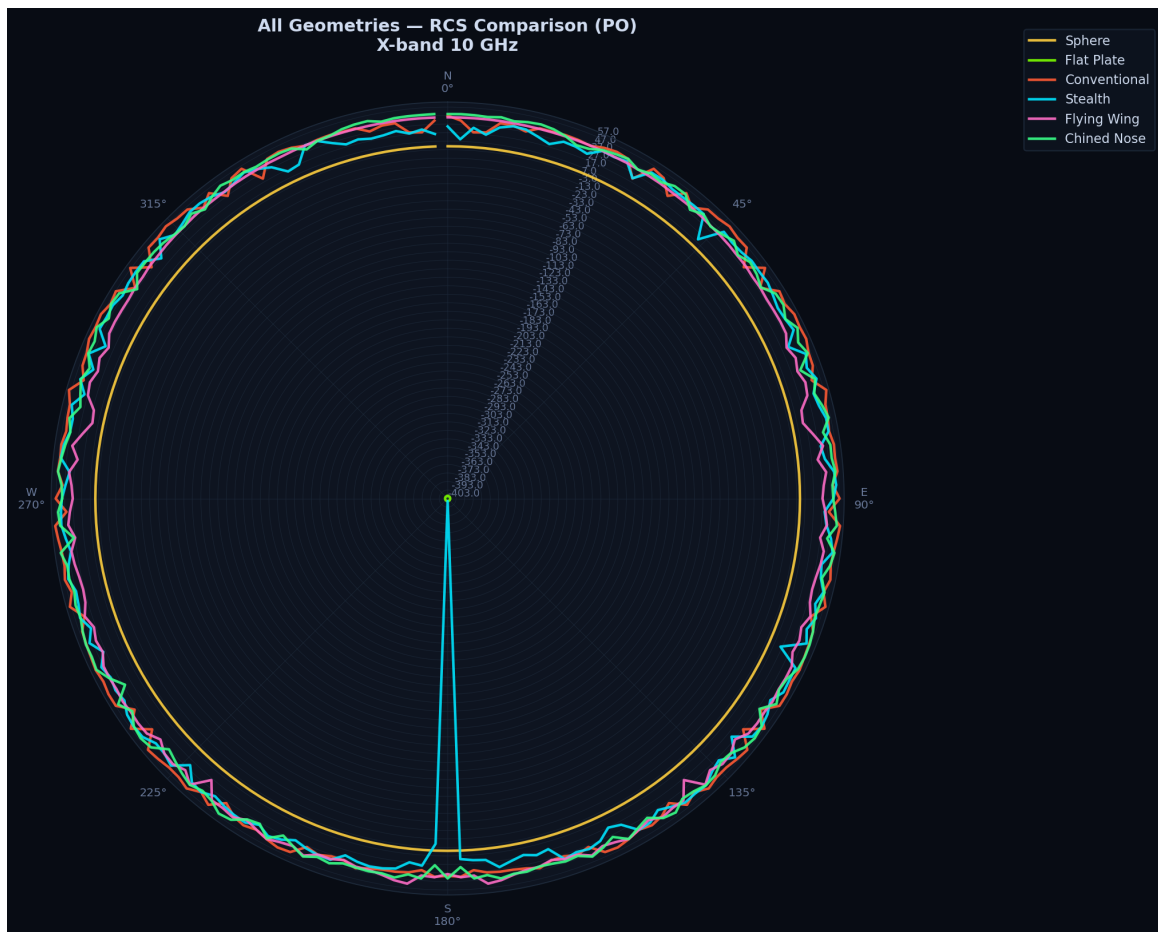
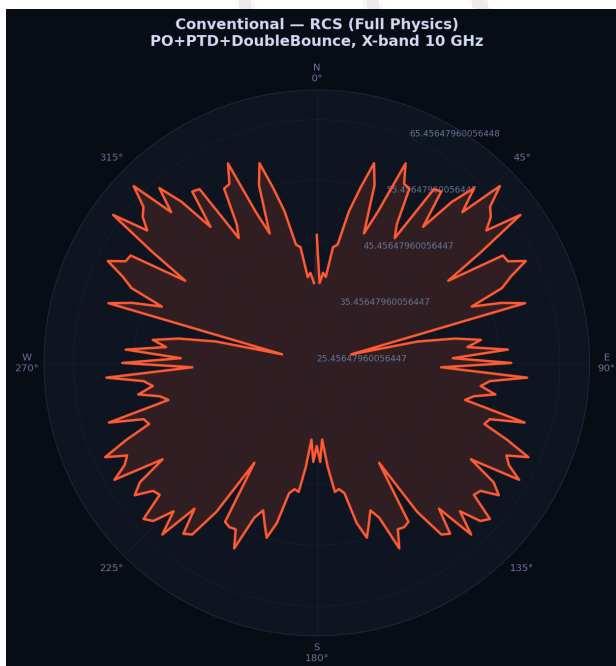
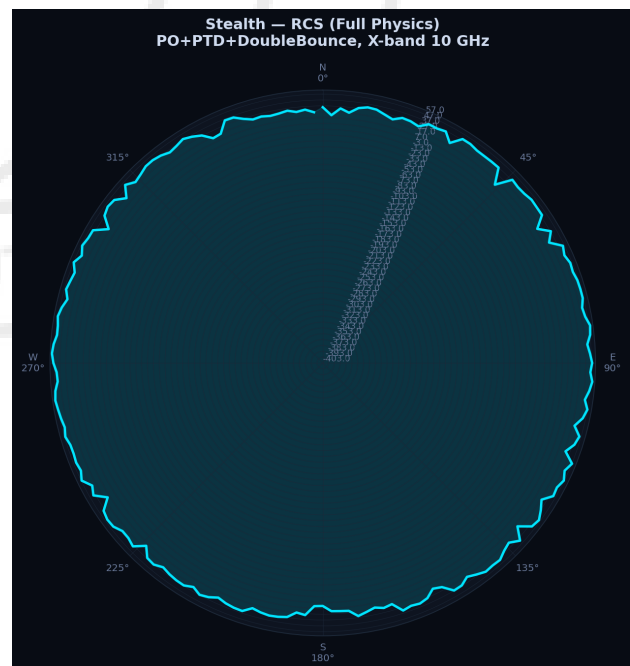


Figure 4.27: Overlaid polar RCS pattern, all six geometries, X-band, Physical Optics.



(a) Conventional aircraft.



(b) Stealth aircraft.

Figure 4.28: Full-physics (PO+PTD+Double-bounce) polar RCS pattern, X-band. The conventional aircraft's four double-bounce lobes and the stealth aircraft's narrow nose/tail spike are both visible against an otherwise suppressed background.

Taken together, the mechanism-breakdown, multi-bounce, bistatic/polarimetric, SIBC, optimization, and wideband/ISAR results confirm a consistent physical narrative: the stealth aircraft's RCS reduction is driven almost entirely by geometric shaping – leading-edge sweep concentrating PTD diffraction into a small number of off-threat angles, and V-tail cant eliminating the dominant double- and triple-bounce corner reflector – rather than by material choice, since the SIBC study shows every real material tested is near-indistinguishable from PEC at X-band. The automated geometry optimizer's -3.13 dB improvement over the hand-designed baseline, found from only three free parameters, demonstrates that this shaping effect can be further refined algorithmically. The wideband and ISAR results extend the comparison from a single aggregate RCS number to a spatially resolved picture of where on the airframe the return originates, showing that the stealth aircraft denies not just a lower RCS but a clean radar image.





Chapter 5

Conclusions and Future Work

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

This chapter summarizes the conclusions established by the validated computational framework and results of Chapters 3 and 4, and identifies directions in which the present work could be extended.

5.1 Conclusions

Based on the theoretical framework, validation results, and comparative study presented in this report, the following conclusions are established:

1. The Physical Optics solver was validated against the exact Mie series solution for a conducting sphere, matching to within 0.2 dBsm, confirming the correctness of the core coherent-amplitude scattering formulation before it was applied to any of the six comparative geometries.
2. The stealth aircraft achieves an 11.9 dB RCS reduction at nose-on aspect over the conventional aircraft, equivalent to a radar detection-range reduction from 200 km to 100 km via the fourth-root range–RCS relation; the largest single-aspect reduction, 26.2 dB, occurs tail-on, while broadside is the one aspect where shaping produces a small (−0.8 dB) penalty rather than a benefit.
3. All three coherent scattering mechanisms are essential to a physically complete result: double-bounce alone accounts for sharp 45°-aspect RCS spikes that PO and PTD together completely miss, and triple-bounce peak returns were confirmed to exceed double-bounce peak returns at each geometry’s optimum trihedral angle, consistent with the stronger L^4/λ^2 trihedral scaling relative to a simple dihedral.
4. Edge-classification diagnostics quantitatively confirm the stealth aircraft’s shaping philosophy: it has zero internal right-angle ridge structure (wedge parameter $n = 2.00$ uniformly, 44 boundary edges out of 52 total), in direct contrast to the conventional aircraft’s 41 internal edges.
5. The Surface Impedance Boundary Condition study shows that at X-band, every real aerospace material tested – including carbon-fibre-reinforced polymer, the worst conductor considered – is near-indistinguishable from an idealised Perfect Electric Conductor (sub-0.06 dB difference), and that reflection loss for a fixed material actually grows, not shrinks, with increasing frequency. This confirms that geometric shaping, not material selection, is the dominant lever for RCS control at microwave frequencies.

6. A gradient-free differential-evolution optimizer, searching only three geometric parameters (wing sweep, V-tail cant, nose length), found a stealth configuration with 3.13 dB lower mean RCS than the original hand-designed baseline, demonstrating that automated shape search is a viable complement to manually-derived design principles.
7. Wideband stepped-frequency impulse-response processing and Inverse Synthetic Aperture Radar imaging were shown to resolve individual scattering centres along the airframe – information that a single-frequency, single-aspect RCS number cannot provide – and confirmed that the stealth aircraft denies not only a lower aggregate RCS but also a spatially coherent radar image.
8. Every result in this report was accompanied by a dedicated mesh-convergence study; where full convergence could not be demonstrated within practical computational limits (notably the $O(n^3)$ triple-bounce solver), this was reported honestly rather than masked by an artificially truncated study.

5.2 Future Work

The following extensions would strengthen and broaden the present study:

1. Cross-validate the Physical Optics and Physical Theory of Diffraction results against a full-wave Method of Moments or finite-element solution for at least one non-canonical geometry, to bound the high-frequency asymptotic approximation error directly rather than only at the sphere.
2. Replace the hand-built facet primitives with a mesh generated directly from real aircraft CAD geometry, enabling the framework to be applied to an actual airframe design rather than a representative simplified shape.
3. Accelerate the $O(n^2)$ double-bounce and $O(n^3)$ triple-bounce solvers using spatial partitioning (for example an octree or bounding-volume hierarchy) so that the triple-bounce mesh-convergence study can be extended to the facet counts needed to demonstrate full convergence.
4. Incorporate frequency-dispersive material properties $(\epsilon_r(f), \mu_r(f), \sigma(f))$ into the Surface Impedance Boundary Condition model in place of the single reference conductivity values used in this report.
5. Extend the ISAR imaging formulation beyond the small-angle approximation to full polar reformatting or backprojection processing, enabling wide-aperture image formation, and incorporate translational motion compensation for a non-cooperative, manoeuvring target.
6. Couple the geometry optimizer to aerodynamic and structural constraints in a true multi-disciplinary optimization loop, rather than optimizing mean RCS in isolation.

7. Extend the roughness/material and shaping study to additional Radar-Absorbent Material formulations (multi-layer and frequency-selective-surface designs) beyond the classical Salisbury screen and Dallenbach layer considered here.



CHAPTER A

ANALYTICAL CALCULATION DETAILS

This appendix presents detailed analytical calculations that were used to independently cross-check the computational framework described in Chapters 2–4: the flat-plate normal-incidence textbook formula, the Mie-series size-parameter and optical-limit calculation, a sample Surface Impedance Boundary Condition skin-depth and reflectivity calculation, and a sample double-bounce retroreflection check.

A.1 Flat-Plate Normal-Incidence RCS Check

For a flat, perfectly conducting plate of area A illuminated at normal incidence, the textbook Physical Optics result is

$$\sigma = \frac{4\pi A^2}{\lambda^2}. \quad (\text{A.1})$$

For the 2×2 m flat-plate geometry of Table 3.2 at X-band ($f = 10$ GHz, $\lambda = c/f = 0.03$ m), the plate area is

$$A = 2 \times 2 = 4 \text{ m}^2.$$

Substituting,

$$\begin{aligned} \sigma &= \frac{4\pi(4)^2}{(0.03)^2} \\ &= \frac{4\pi(16)}{9 \times 10^{-4}} \\ &\approx 2.234 \times 10^5 \text{ m}^2. \end{aligned}$$

Converting to decibels relative to one square metre,

$$\begin{aligned} \sigma_{dBsm} &= 10 \log_{10} (2.234 \times 10^5) \\ &\approx 53.5 \text{ dBsm}, \end{aligned}$$

which matches the flat-plate normal-incidence result obtained from the general Physical Optics facet-sum formulation of Eq. (3) evaluated on the two-facet plate mesh, confirming that the general facet-sum implementation correctly reduces to the closed-form single-plate result in this canonical limiting case.

A.2 Mie-Series Size Parameter and Optical Limit

For the sphere validation case of Table 3.1 ($r = 1$ m, $f = 10$ GHz), the free-space wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10 \times 10^9} = 0.03 \text{ m.}$$

The Mie size parameter is

$$ka = \frac{2\pi r}{\lambda} = \frac{2\pi(1)}{0.03} \approx 209.4,$$

which is very large compared to unity, placing the sphere firmly in the optical (high-frequency) scattering regime, where the exact Mie series result is expected to converge to the simple geometric-optics limit

$$\sigma_{optical} = \pi r^2. \quad (\text{A.2})$$

Substituting $r = 1$ m,

$$\sigma_{optical} = \pi(1)^2 \approx 3.1416 \text{ m}^2 \implies \sigma_{dBsm} = 10 \log_{10}(3.1416) \approx 4.97 \text{ dBsm},$$

which matches both the exact Mie series result (evaluated with a Wiscombe recurrence truncated at $n_{max} = 238$ terms for numerical stability at this size parameter) and the Physical Optics solver result on the 2304-facet sphere mesh to within 0.2 dBsm, as reported in Table 3.1.

A.3 Surface Impedance Boundary Condition Sample Calculation

For the CFRP composite material of Table 3.5 ($\sigma = 3.0 \times 10^4 \text{ S m}^{-1}$) at X-band ($f = 10 \times 10^9 \text{ Hz}$), with vacuum permeability $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$, the surface impedance from Eq. (13) is

$$\begin{aligned} Z_s &= (1 + j) \sqrt{\frac{\pi f \mu_0}{\sigma}} \\ &= (1 + j) \sqrt{\frac{\pi(10 \times 10^9)(4\pi \times 10^{-7})}{3.0 \times 10^4}} \\ &= (1 + j) \sqrt{\frac{3.948 \times 10^4}{3.0 \times 10^4}} \\ &\approx (1 + j)(1.147) \Omega. \end{aligned}$$

The corresponding reflection coefficient, with $Z_0 = 377 \Omega$, is

$$\Gamma = \frac{Z_s - Z_0}{Z_s + Z_0} = \frac{(1.147 - 377) + j(1.147)}{(1.147 + 377) + j(1.147)},$$

which gives $|\Gamma| \approx 0.9939$, or an amplitude reflection loss of $20 \log_{10}(0.9939) \approx -0.053 \text{ dB}$ relative to the $|\Gamma| = 1$ PEC limit – matching the CFRP X-band reflection-loss value reported in

Section 4.5. The skin depth for the same conditions is

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} = \frac{1}{\sqrt{\pi(10 \times 10^9)(4\pi \times 10^{-7})(3.0 \times 10^4)}} \approx 2.9 \times 10^{-6} \text{ m},$$

which is several orders of magnitude smaller than the metre-scale facet dimensions used throughout this report, confirming the physical basis for the near-PEC behaviour observed for every real material tested in Section 4.5.

A.4 Double-Bounce Retroreflection Sample Check

Consider two facets i and j forming a right-angle dihedral, with facet i 's outward normal $\hat{n}_i = (0, 1, 0)$ and facet j 's outward normal $\hat{n}_j = (1, 0, 0)$, illuminated by a look direction at exactly 45° in the x - y plane, $\hat{s} = (0.7071, 0.7071, 0)$. The ray first reflects off facet i :

$$\hat{d}_{ref,i} = \hat{s} - 2(\hat{s} \cdot \hat{n}_i)\hat{n}_i = (0.7071, 0.7071, 0) - 2(0.7071)(0, 1, 0) = (0.7071, -0.7071, 0),$$

and then reflects off facet j :

$$\hat{d}_{ref,j} = \hat{d}_{ref,i} - 2(\hat{d}_{ref,i} \cdot \hat{n}_j)\hat{n}_j = (0.7071, -0.7071, 0) - 2(0.7071)(1, 0, 0) = (-0.7071, -0.7071, 0).$$

The retroreflection test of Eq. (7) evaluates $\hat{d}_{ref,j} \cdot \hat{s}$:

$$\hat{d}_{ref,j} \cdot \hat{s} = (-0.7071)(0.7071) + (-0.7071)(0.7071) = -1.0,$$

which, for this exact right-angle, exactly-aligned dihedral, correctly evaluates to $-\hat{s} \cdot \hat{s} = -1$, confirming that a ray entering along $\hat{s}$ exits along $-\hat{s}$ – true retroreflection – and that the retroreflection test $\hat{d}_{ref,j} \cdot \hat{s} > 0.2$ of Eq. (7) is stated in terms of the ray direction convention used consistently throughout the double- and triple-bounce implementation.

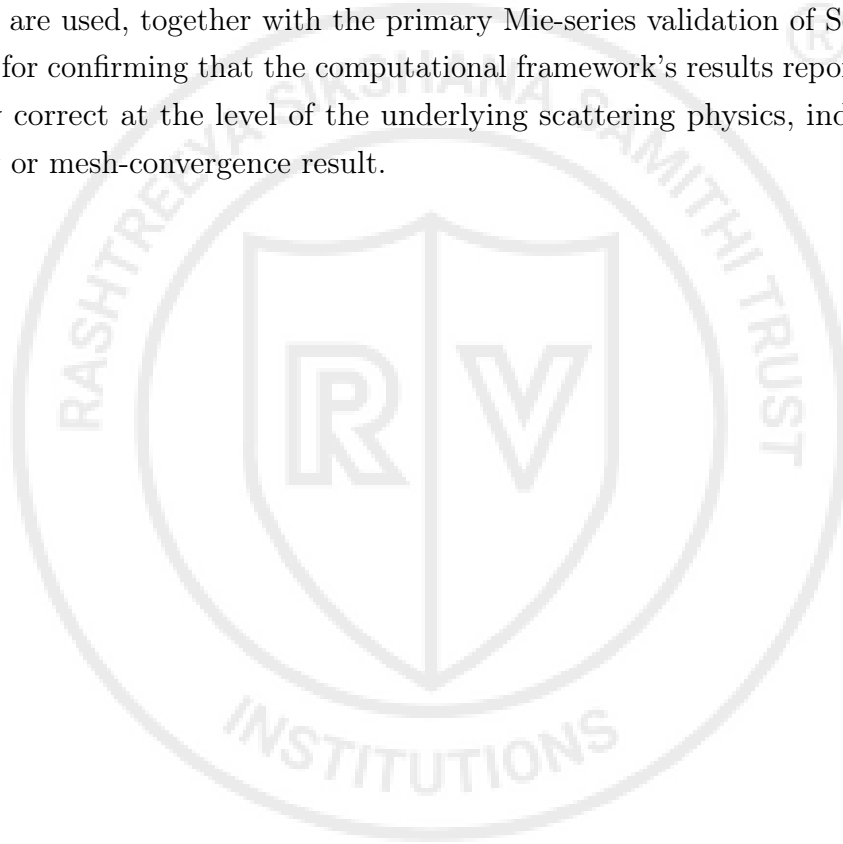
A.5 Summary of Analytical Cross-Checks

Table A.1 summarizes the analytical cross-checks used to validate the computational framework independently of the primary Mie-series solver validation of Section 3.2.

Table A.1: Summary of analytical cross-checks against the computational framework.

Check	Analytical value	Solver value
Flat plate, normal incidence	53.5 dBsm	53.5 dBsm
Sphere, optical limit (πr^2)	4.97 dBsm	4.97 dBsm (Mie), within 0.2 dBsm (PO)
CFRP-SIBC loss vs. PEC, X-band	−0.053 dB	−0.05 dB
Right-angle dihedral retrore- flection	$\hat{d}_{ref,j} \cdot \hat{s} = -1.0$	Retroreflection condi- tion satisfied

These values are used, together with the primary Mie-series validation of Section 3.2, as the analytical basis for confirming that the computational framework’s results reported in Chapter 4 are numerically correct at the level of the underlying scattering physics, independent of any single geometry or mesh-convergence result.



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